

Electromagnetic Fields and Transmission Lines

Previous Year Questions • 1987–2026

GATE ELECTRONICS AND COMMUNICATION ENGINEERING

(EC)

Himanshu Agarwal & Anish Saha

PrepFusion™

Where Concept Meets Clarity!

Preface

Welcome to PrepFusion™!

This book is a complete compilation of the questions GATE has actually asked in Electromagnetic Fields and Transmission Lines (EC), gathered from the original papers and arranged unit by unit and year by year. Nothing is paraphrased or reconstructed: every question appears as it was set, tagged with its year, session and marks.

Each question carries an identifier that maps one-to-one onto its worked solution in the companion solutions volume, so you can move between practice and explanation without a lookup table. Answer keys close every section, and answers revised by the examining authority (MTA) are marked where they occur.

Used end to end, the book shows you what the paper rewards — which topics repeat, how the framing has shifted over the years, and where your preparation still has gaps.

Meet your Authors

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> *"We built this book the way every aspirant deserves one to be built — organized strictly by unit and by year, so that every hour of practice maps directly onto how GATE actually tests you."*
>
> — Himanshu Agarwal & Anish Saha

How to Read the Question Numbers

Every question in this book carries a three-part identifier instead of a plain serial number:

Q3.24.7

3	Unit (chapter) number — Unit III of this book
24	GATE exam year — 2024 (two digits; 98 = 1998)
7	Question number within that unit and year

So **Q3.24.7** is the 7th question that GATE 2024 contributed to Unit III. Numbers are therefore not sequential through the book — they tell you the unit and the exam year at a glance. The same identifier is used in the questions book, the solutions book and the answer keys, so you can jump between them without a lookup table. In the PDF bookmarks panel, expand a unit to see every question ID and click one to go straight to it.

What the Bubbles Mean

Every question ends with three empty circles for you to shade in yourself:

○○○ ●○○ Easy ●●○ Medium ●●● Hard

The companion solutions volume marks the same question with *our* recommended difficulty, using the identical bubbles. If you think a rating should be different, tell us directly — report it under the **Study Hub** section at **pyq.prepfusion.in**.

How to Read the Unit Snapshot

Every unit opens with a **Unit Snapshot** box summarizing that unit's real question data:

Questions	the real total for this unit.
MCQ, MSQ, NAT	that same total, split by question type.
1M, 2M	how many of those questions carry 1 or 2 marks.
Descriptive	questions worth more than 2 marks, or whose marks value isn't recorded.
Avg Weightage	the average marks this unit has contributed per real GATE sitting, not a raw year-by-year total — a year GATE ran multiple real sessions (a genuine forenoon/afternoon split, not the same paper's Set A/B/C/D booklet variants) is divided across those sessions first, so a heavily multi-session year cannot silently outweigh a single-session one. The ~ marks it as a rounded figure, not an exact one.
Range	the lowest and highest that same per-sitting value has been, with the year each occurred.

Avg Weightage and Range are computed from **2010 onward** only: GATE dropped from a 150-mark to a 100-mark paper in 2009, but that year itself was a one-off transition (60 questions, not the 65-question format that became standard from 2010 on) — so 2010 is the first year fully comparable to modern papers, and marks-based figures from earlier years are left out of these two figures specifically (every other count on this page still includes every year). They also depend on knowing which GATE years ran more than one real sitting for a given branch — our own research into this is thorough for some branches and still a work in progress for others. Treat both figures as a **beta**, best-effort estimate: directionally useful, not exact.

MSQ questions specifically weren't introduced until 2021 — an older-year-heavy unit showing **MSQ: 0** reflects that real exam history, not a data gap.

Worked example

Unit Snapshot*

Stream: EC	**Questions:** 45	**MCQ:** 30	**MSQ:** 3	**NAT:** 12
1M: 20	**2M:** 22	**Descriptive:** 3	**Avg Weightage** ~4M	**Range:** 2M(2015)–6M(2021)

45 real questions total; on average this unit has been worth about 4 marks per real GATE sitting since 2010, ranging from as low as 2 (in 2015) to as high as 6 (in 2021).

A quick way to use this: a high **Avg Weightage** points to a consistently high-yield topic worth prioritizing; a wide **Range** usually means the topic's importance has shifted over the years, so it's worth checking *which* years drove that before deciding how much weight to give it today.

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Scan any code with your phone camera, or tap the link beneath it. **pyq.prepfusion.in** is also where you can read this book's worked solutions online — and in its **Study Hub** section, report an error in any question or request a video solution for it.

Contributors

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UNIT

I

Coordinate Geometry & Maxwell's Equation - Basics

Topics

1. Basics
2. Maxwell's Equation for Static & Time Varying Fields

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Unit Snapshot*

Stream: EC	**Questions:** 32	**MCQ:** 23	**MSQ:** 1	**NAT:** 8
1M: 9	**2M:** 23	**Descriptive:** 0	**Avg Weightage** ~2M	**Range:** 1M(2019)–6M(2025)

**See preface for how Avg Weightage and Range are calculated.*

Q1.25.1 MCQ 2025 2M

A square metal sheet of 4 m × 4 m is placed on the x-y plane as shown in the figure below.

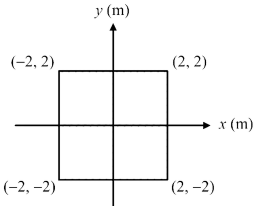


Fig. Q1.25.1

If the surface charge density (in μC/m^2) on the sheet is $\rho_s(x, y) = 4|y|$, then the total charge (in μC, rounded off to the nearest integer) on the sheet is _____.

A. 16 **B.** 85 **C.** 64 **D.** 256

Q1.25.2 MCQ 2025 2M

An electric field of 0.01 V/m is applied along the length of a copper wire of circular cross-section with diameter 1 mm. Copper has a conductivity of 5.8×10^7 S/m.
The current (in Amperes, rounded off to two decimal places) flowing through the wire is _____.

A. 0.46 **B.** 1.82 **C.** 0.58 **D.** 1.12

Q1.25.3 NAT 2025 2M

Two resistors are connected in a circuit loop of area 5 m^2, as shown in the figure below. The circuit loop is placed on the x-y plane.

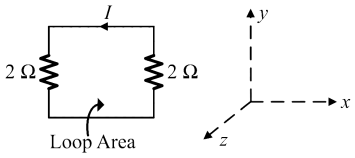


Fig. Q1.25.3

When a time-varying magnetic flux, with flux-density $B(t) = 0.5t$ (in Tesla), is applied along the positive z-axis, the magnitude of current I (in Amperes, rounded off to two decimal places) in the loop is _____.

Q1.24.1 MCQ 2024 2M

Consider the Earth to be a perfect sphere of radius R. Then the surface area of the region, enclosed by the 60°N latitude circle, that contains the north pole in its interior is _____.

A. $(2 - \sqrt{3})\pi R^2$ **B.** $\dfrac{(\sqrt{2} - 1)\pi R^2}{2}$

C. $\dfrac{2\pi R^2}{3}$ **D.** $\dfrac{(2 + \sqrt{3})\pi R^2}{8\sqrt{2}}$

Q1.24.2 MCQ 2024 1M

Let $\hat{i}$ and $\hat{j}$ be the unit vectors along x and y axes, respectively and let A be a positive constant. Which one of the following statements is true for the vector fields $\vec{F}_1 = A(\hat{i}y + \hat{j}x)$ and $\vec{F}_2 = A(\hat{i}y - \hat{j}x)$?

A. Both $\vec{F}_1$ and $\vec{F}_2$ are electrostatic fields.

B. Only $\vec{F}_1$ is an electrostatic field.

C. Only $\vec{F}_2$ is an electrostatic field.

D. Neither $\vec{F}_1$ nor $\vec{F}_2$ is an electrostatic field.

Q1.22.1 NAT 2022 2M

In an electrostatic field, the electric displacement density vector, $\vec{D}$, is given by

$$\vec{D}(x, y, z) = (x^3\vec{i} + y^3\vec{j} + xy^2\vec{k})\ \text{C/m}^2,$$

where $\vec{i}$, $\vec{j}$, $\vec{k}$ are the unit vectors along x-axis, y-axis, and z-axis, respectively. Consider a cubical region R centered at the origin with each side of length 1 m, and vertices at (±0.5 m, ±0.5 m, ±0.5 m). The electric charge enclosed within R is _____ C (rounded off to two decimal places).

Q1.22.2 MCQ 2022 1M

In a circuit, there is a series connection of an ideal resistor and an ideal capacitor. The conduction current (in Amperes) through the resistor is $2\sin(t + \pi/2)$. The displacement current (in Amperes) through the capacitor is _____.

A. $2\sin(t)$

B. $2\sin(t + \pi)$

C. $2\sin(t + \pi/2)$

D. 0

Q1.21.1 NAT 2021 2M

For a vector field $\mathbf{D} = \rho\cos^2\varphi\mathbf{a}_\rho + z^2\sin^2\varphi\mathbf{a}_\varphi$ in a cylindrical coordinate system (ρ, φ, z) with unit vectors $\mathbf{a}_\rho$, $\mathbf{a}_\varphi$ and $\mathbf{a}_z$, the net flux of $\mathbf{D}$ leaving the closed surface of the cylinder $(\rho = 3, 0 \le z \le 2)$ (*rounded off to two decimal places*) is _____.

Q1.21.2 NAT 2021 1M

Consider the vector field $\mathbf{F} = \mathbf{a}_x(4y - c_1z) + \mathbf{a}_y(4x + 2z) + \mathbf{a}_z(2y + z)$ in a rectangular coordinate system (x, y, z) with unit vectors $\mathbf{a}_x$, $\mathbf{a}_y$, and $\mathbf{a}_z$. If the field $\mathbf{F}$ is irrotational (conservative), then the constant c_1 (*in integer*) is _____.

Q1.19.1 MCQ 2019 1M

In the table shown, **List-I** and **List-II** respectively, contain terms appearing on the left-hand side and the right-hand side of Maxwell's equations (in their standard form). Match the left-hand side withe the corresponding right-hand side.

List-I	List-II
1. $\nabla \cdot \vec{D}$	P. 0
2. $\nabla \times \vec{E}$	Q. ρ
3. $\nabla \cdot \vec{B}$	R. $-\dfrac{\partial \vec{B}}{\partial t}$
4. $\nabla \times \vec{H}$	S. $\vec{J} + \dfrac{\partial \vec{D}}{\partial t}$

A. 1 - R, 2 - Q, 3 - S, 4 - P

B. 1 - Q, 2 - S, 3 - P, 4 - R

C. 1 - P, 2 - R, 3 - Q, 4 - S

D. 1 - Q, 2 - R, 3 - P, 4 - S

Q1.16.1 MCQ 2016 2M

The parallel-plate capacitor shown in the figure has movable plates. The capacitor is charged so that the energy stored in it is E when the plate separation is d. The capacitor is then isolated electrically and the plates are moved such that the plate separation becomes $2d$.

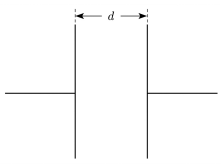


Fig. Q1.16.1

At this new plate separation, what is the energy stored in the capacitor, neglecting fringing effect?

A. $2E$ **B.** $\sqrt{2}E$
C. E **D.** $E/2$

Q1.16.2 MCQ 2016 2M

The current density in a medium is given by

$$\vec{J} = \frac{400\sin\theta}{2\pi(r^2+4)}\hat{a}_r,\ \text{Am}^{-2}$$

The total current and the average current density flowing through the portion of a spherical surface, $r =$ 0.8 m, $\dfrac{\pi}{12} \le \theta \le \dfrac{\pi}{4}$, $0 \le \phi \le 2\pi$ are given, respectively by

A. 15.09 A, 12.86 Am^{-2}

B. 18.73 A, 13.65 Am^{-2}

C. 12.86 A, 9.23 Am^{-2}

D. 10.28 A, 7.56 Am^{-2}

Q1.16.3 NAT 2016 1M

Concentric spherical shells of radii 2 m, 4 m and 8 m carry uniform surface charge densities of 20 nC/m^2, -4 nC/m^2 and ρ_s, respectively. The value of ρ_s (nC/m^2) required to ensure that the electric flux density $\vec{D} = \vec{0}$ at radius 10 m is _____.

Q1.15.1 NAT 2015 2M

A vector field $\vec{D} = 2\rho^2\hat{a}_\rho + z\hat{a}_z$ exists inside a cylindrical region enclosed by the surfaces $\rho = 1$, $z = 0$ and $z = 5$. Let S be the surface bounding this cylindrical region. The surface integral of this field on S $\oiint_S \vec{D} \cdot d\vec{s}$ is _____.

Q1.15.2 NAT 2015 1M

In a source free region in vacuum, if the electrostatic potential $\phi = 2x^2 + y^2 + cz^2$, the value of constant c must be _____.

Q1.14.1 MCQ 2014 2M

If $\vec{E} = -(2y^3 - 3yz^2)\hat{x} - (6xy^2 - 3xz^2)\hat{y} + (6xyz)\hat{z}$ is the electric field in a source free region, a valid expression for the electrostatic potential is

A. $xy^3 - yz^2$ **B.** $2xy^3 - xyz^2$
C. $y^3 + xyz^2$ **D.** $2xy^3 - 3xyz^2$

Q1.14.2 MCQ 2014 2M

A region shown below contains a perfect conducting half-space and air. The surface current $\vec{K}_s$ on the surface of the perfect conductor is $\vec{K}_s = \hat{x}2$ amperes per meter. The tangential $\vec{H}$ field in the air just above the perfect conductor is

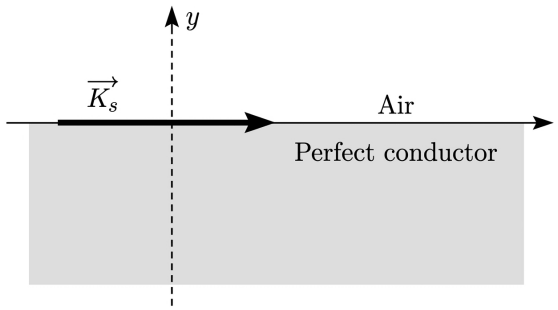


Fig. Q1.14.2

A. $(\hat{x} + \hat{z})$ amperes per meter

B. $\hat{x}2$ amperes per meter

C. $-\hat{z}2$ amperes per meter

D. $\hat{z}2$ amperes per meter

Q1.11.1 MCQ 2011 2M

A current sheet $\vec{J} = 10\hat{u}_y$ A/m lies on the dielectric interface $x = 0$ between two dielectric media with $\varepsilon_{r1} = 1, \mu_{r1} = 1$ in Region-1 ($x < 0$) and $\varepsilon_{r2} = 2, \mu_{r2} = 2$ in Region-2 ($x > 0$). If the magnetic field in Region-1 at $x = 0^-$ is $\vec{H}_1 = 3\hat{u}_x + 30\hat{u}_y$ A/m, the magnetic field in Region-2 at $x = 0^+$ is

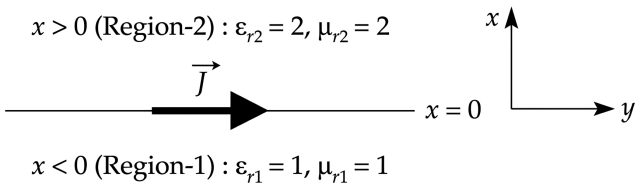


Fig. Q1.11.1

A. $\vec{H}_2 = 1.5\hat{u}_x + 30\hat{u}_y - 10\hat{u}_z$ A/m

B. $\vec{H}_2 = 3\hat{u}_x + 30\hat{u}_y - 10\hat{u}_z$ A/m

C. $\vec{H}_2 = 1.5\hat{u}_x + 40\hat{u}_y$ A/m

D. $\vec{H}_2 = 3\hat{u}_x + 30\hat{u}_y + 10\hat{u}_z$ A/m

Q1.09.1 MCQ 2009 2M

If a vector field $\vec{V}$ is related to another vector field $\vec{A}$ through $\vec{V} = \nabla \times \vec{A}$, which of the following is true?

Note: C and S_C refer to any closed contour and any surface whose boundary is C.

A. $\oint_C \vec{V} \cdot \vec{dl} = \iint_{S_C} \vec{A} \cdot \vec{ds}$

B. $\oint_C \vec{A} \cdot \vec{dl} = \iint_{S_C} \vec{V} \cdot \vec{ds}$

C. $\oint_C \nabla \times \vec{V} \cdot \vec{dl} = \iint_{S_C} \nabla \times \vec{A} \cdot \vec{ds}$

D. $\oint_C \nabla \times \vec{A} \cdot \vec{dl} = \iint_{S_C} \vec{V} \cdot \vec{ds}$

Q1.08.1 MCQ 2008 2M

For static electric and magnetic fields in an inhomogeneous source free medium, which of the following represents the correct form of two of Maxwell's equations?

A. $\nabla \cdot \vec{E} = 0$, $\nabla \times \vec{B} = 0$

B. $\nabla \cdot \vec{E} = 0$, $\nabla \cdot \vec{B} = 0$

C. $\nabla \times \vec{E} = 0$, $\nabla \times \vec{B} = 0$

D. $\nabla \times \vec{E} = 0$, $\nabla \cdot \vec{B} = 0$

Q1.07.1 MCQ 2007 1M

If C is a closed curve enclosing a surface S, then the magnetic field intensity $\vec{H}$, the current density $\vec{J}$ and the electric flux density $\vec{D}$ are related by

A. $\iint_S \vec{H} \cdot \vec{ds} = \oint_C \left(\vec{J} + \frac{\partial \vec{D}}{\partial t}\right) \cdot \vec{dl}$

B. $\int_C \vec{H} \cdot \vec{dl} = \oiint_S \left(\vec{J} + \frac{\partial \vec{D}}{\partial t}\right) \cdot \vec{ds}$

C. $\oiint_S \vec{H} \cdot \vec{ds} = \int_C \left(\vec{J} + \frac{\partial \vec{D}}{\partial t}\right) \cdot \vec{dl}$

D. $\oint_C \vec{H} \cdot \vec{dl} = \iint_S \left(\vec{J} + \frac{\partial \vec{D}}{\partial t}\right) \cdot \vec{ds}$

Q1.06.1 MCQ 2006 2M

A medium is divided into regions I and II about $x = 0$ plane, as shown in the figure below. An electromagnetic wave with electric field $\vec{E}_1 = 4\hat{a}_x + 3\hat{a}_y + 5\hat{a}_z$ is incident normally on the interface from region-I. The electric field $\vec{E}_2$ in region-II at the interface is

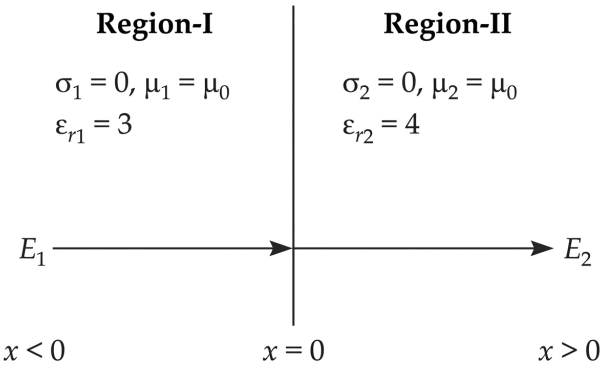


Fig. Q1.06.1

A. $\vec{E}_2 = \vec{E}_1$
B. $4\hat{a}_x + 0.75\hat{a}_y - 1.25\hat{a}_z$
C. $3\hat{a}_x + 3\hat{a}_y + 5\hat{a}_z$
D. $-3\hat{a}_x + 3\hat{a}_y + 5\hat{a}_z$

Q1.03.1 MCQ 2003 2M

Medium 1 has the electrical permittivity $\epsilon_1 = 1.5\epsilon_0$ Farad/m and occupies the region to the left of $x = 0$ plane. Medium 2 has the electrical permittivity $\epsilon_2 = 2.5\epsilon_0$ Farad/m and occupies the region to the right of $x = 0$ plane. If E_1 in medium 1 is $E_1(2u_x - 3u_y + 1u_z)$ Volt/m, then E_2 in medium 2 is

A. $(2.0u_x - 7.5u_y + 2.5u_z)$ Volt/m
B. $(2.0u_x - 2.0u_y + 0.6u_z)$ Volt/m
C. $(1.2u_x - 3.0u_y + 1.0u_z)$ Volt/m
D. $(1.2u_x - 2.0u_y + 0.6u_z)$ Volt/m

Q1.03.2 MCQ 2003 2M

If the electric field intensity is given by

$$\vec{E} = (x\hat{u}_x + y\hat{u}_y + z\hat{u}_z) \text{ Volt/m}$$

The potential difference between $X(2, 0, 0)$ and $Y(1, 2, 3)$ is

A. +1 Volt
B. -1 Volt
C. +5 Volt
D. +6 Volt

Q1.03.3 MCQ 2003 1M

The unit of $\nabla \times \vec{H}$ is

A. Ampere
B. Ampere/meter
C. Ampere/meter^2
D. Ampere-meter

Q1.02.1 MCQ 2002 2M

The electric field on the surface of a perfect conductor is 2 V/m. The conductor is immersed in water with $\epsilon = 80\epsilon_0$. The surface charge density on the conductor is

A. 0 C/m^2
B. 2 C/m^2
C. 1.8×10^{-11} C/m^2
D. 1.41×10^{-9} C/m^2

Q1.96.1 MCQ 1996 1M

A metal sphere with 1 m radius and a surface charge density of 10 C/m^2 is enclosed in a cube of 10 m side. The total outward electric displacement flux normal to the surface of the cube is

A. 40π Coulombs
B. 10π Coulombs
C. 5π Coulombs
D. None of the above

Q1.93.1 NAT 1993 2M

Given, $\vec{V} = x\cos^2 y\,\hat{i} + x^2 e^z\,\hat{j} + z\sin^2 y\,\hat{k}$ and S the surface of unit cube with one corner at the origin and edges parallel to the coordinate axis, the value of the integral $\iint_C \vec{V} \cdot \hat{n}\, dS$ is _____.

Q1.90.1 MCQ 1990 2M

Which of the following field equations indicate that the free magnetic charge do not exist?

A. $\vec{H} = \frac{1}{\mu}\nabla \times \vec{A}$
B. $\vec{H} = \oint \frac{Id\vec{l} \times \vec{R}}{4\pi R^2}$
C. $\nabla \cdot \vec{H} = 0$
D. $\nabla \times \vec{H} = \vec{J}$

Q1.88.1 MCQ 1988 2M

Vector potential is a vector

A. whose curl is equal to the magnetic flux density.
B. whose curl is equal to the electric field intensity.
C. whose divergence is equal to the electric potential.
D. which is equal to the vector product $\vec{E} \times \vec{H}$.

Q1.88.2 MSQ 1988 2M

On either side of a charge free interface between two media

☐ **A.** the normal components of the electric field are equal.
☐ **B.** the tangential components of the electric field are equal.

☐ **C.** the normal components of the electric flux density are equal.

☐ **D.** the tangential components of the electric flux density are equal.

Q1.87.1 MCQ 1987 2M ◯◯◯

An electrostatic field is said to be conservative when:

A. The divergence of the field is equal to zero.

B. The curl of the field is equal to zero.

C. The curl of the field is equal to $-\frac{\partial E}{\partial t^2}$.

D. The Laplacian of the field is equal to $\mu\varepsilon\frac{\partial^2 E}{\partial t^2}$.

Answer Key

Note: **MTA** Marks To All awarded — see detailed solutions.

Q1.25.1 C	**Q1.25.2** A	**Q1.25.3** 0.62 to 0.63
Q1.24.1 A	**Q1.24.2** B	**Q1.22.1** 0.50
Q1.22.2 C	**Q1.21.1** 56.55	**Q1.21.2** 0
Q1.19.1 D	**Q1.16.1** A	**Q1.16.2** MTA
Q1.16.3 -0.28 to -0.22	**Q1.15.1** 78.5 to 78.6	**Q1.15.2** -3
Q1.14.1 D	**Q1.14.2** D	**Q1.11.1** A
Q1.09.1 B	**Q1.08.1** D	**Q1.07.1** D
Q1.06.1 C	**Q1.03.1** C	**Q1.03.2** C
Q1.03.3 C	**Q1.02.1** D	**Q1.96.1** A
Q1.93.1 1	**Q1.90.1** C	**Q1.88.1** A
Q1.88.2 B,C	**Q1.87.1** B	

UNIT

II

EM Waves

Topics

1. EM Wave Propagation in Different Medium
2. Poynting Vector & Poynting Theorem
3. Boundary Conditions
4. Normal & Oblige Incidence of EM Wave
5. Wave Polarization

PrepFusion

Unit Snapshot*

Stream: EC	**Questions:** 71	**MCQ:** 60	**MSQ:** 4	**NAT:** 7
1M: 31	**2M:** 40	**Descriptive:** 0	**Avg Weightage** ~2M	**Range:** 1M(2021)–4M(2018)

**See preface for how Avg Weightage and Range are calculated.*

Q2.26.1 MCQ 2026 1M

The electric field of a monochromatic plane wave travelling in a lossless isotropic and homogeneous medium is given by

$$\vec{E}(z,t) = E_0[\hat{x}\cos(\omega t - kz) + \hat{y}\sin(\omega t - kz)]$$

in a right-handed orthogonal co-ordinate system. Which of the following is the correct polarization of the electromagnetic wave?

A. Right-handed circularly polarized

B. Left-handed circularly polarized

C. Linearly polarized

D. Linearly polarized with $-45°$ angle to $\hat{x}$

Q2.26.2 MSQ 2026 1M

Which option(s) represents/ represent the dielectric loss tangent of a substrate?

☐ **A.** Ratio of the real to imaginary parts of the total displacement current

☐ **B.** $(\omega\epsilon'' + \sigma)/(\omega\epsilon')$

☐ **C.** Ratio of the electric susceptibility to permittivity

☐ **D.** Ratio of the polarization vector $\vec{P}$ to the displacement vector $\vec{D}$

Q2.24.1 MCQ 2024 2M

A uniform plane wave with electric field $\vec{E}(x) = A_y\hat{a}_y e^{-j\frac{2\pi x}{3}}$ V/m is travelling in the air (relative permittivity, $\epsilon_r = 1$ and relative permeability, $\mu_r = 1$) in the $+x$ direction (A_y is a positive constant, $\hat{a}_y$ is the unit vector along the y axis). It is incident normally on an ideal electric conductor (conductivity, $\sigma = \infty$) at $x = 0$. The position of the first null of the total magnetic field in the air (measured from $x = 0$, in metres) is _____.

A. -6 **B.** $-\frac{3}{2}$

C. -3 **D.** $-\frac{3}{4}$

Q2.23.1 MSQ 2023 2M

The electric field of a plane electromagnetic wave is $E = a_x C_{1x}\cos(\omega t - \beta z) + a_y C_{1y}\cos(\omega t - \beta z + \theta)$ V/m Which of the following combination(s) will give rise to a left handed elliptically polarized (LHEP) wave?

☐ **A.** $C_{1x} = 1, C_{1y} = 1, \theta = \pi/4$

☐ **B.** $C_{1x} = 2, C_{1y} = 1, \theta = \pi/2$

☐ **C.** $C_{1x} = 1, C_{1y} = 2, \theta = 3\pi/2$

☐ **D.** $C_{1x} = 2, C_{1y} = 1, \theta = 3\pi/4$

Q2.21.1 MCQ 2021 1M

Consider a rectangular coordinate system (x, y, z) with unit vectors a_x, a_y and a_z. A plane wave traveling in the region $z \geq 0$ with electric field vector, $E = 10\cos(2\times10^8 t + \beta z)a_y$ is incident normally on the plane at $z = 0$, where β is the phase constant. The region $z \geq 0$ is in free space and the region $z < 0$ is filled with a lossless medium (permittivity, $\varepsilon = \varepsilon_0$, permeability, $\mu = 4\mu_0$, where $\varepsilon_0 = 8.85\times10^{-12}$ F/m and $\mu_0 = 4\pi\times10^{-7}$ H/m). The value of the reflection coefficient is,

A. $\frac{3}{5}$ **B.** $\frac{2}{3}$

C. $\frac{2}{5}$ **D.** $\frac{1}{3}$

Q2.20.1 NAT 2020 2M

The magnetic field of a uniform plane wave in vacuum is given by

$$\vec{H}(x,y,z,t) = (\hat{a}_x + 2\hat{a}_y + b\hat{a}_z)\cos(\omega t + 3x - y - z)$$

The value of b is _____.

Q2.18.1 MCQ 2018 2M

The distance (in m) a wave has to propagate in a medium having a skin depth of 0.1 m so that the amplitude of the wave attenuates by 20 dB, is

A. 0.12 **B.** 0.23

C. 0.46 **D.** 2.3

Q2.18.2 NAT 2018 2M

A uniform plane wave travelling in free space and having the electric field,

$$\vec{E} = (\sqrt{2}\hat{a}_x - \hat{a}_z)\cos[6\sqrt{3}\pi \times 10^8 t - 2\pi(x + \sqrt{2}z)] \text{ V/m}$$

is incident on a dielectric medium (relative permittivity > 1, relative permeability $= 1$) as shown in the figure and there is no reflected wave.

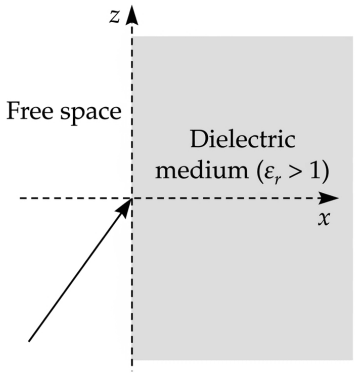


Fig. Q2.18.2

The relative permittivity (correct to two decimal places) of the dielectric medium is _____.

Q2.17.1 NAT 2017 2M

The permittivity of water at optical frequencies is $1.75\varepsilon_0$. It is found that an isotropic light source at a distance 'd' under water forms an illuminated circular area of radius 5 m as shown in the figure. The critical angle is θ_c.

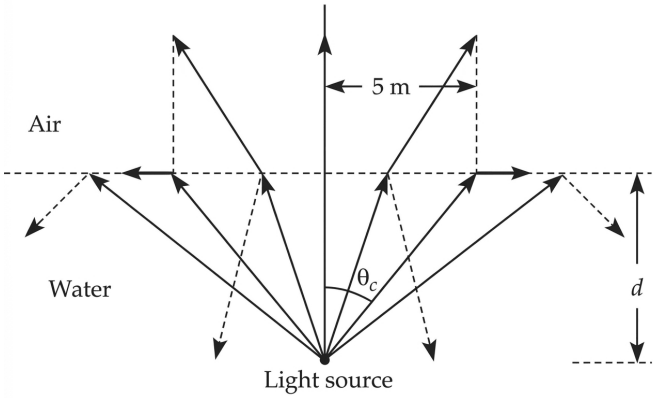


Fig. Q2.17.1

The value of 'd' (in meter) is _____.

Q2.17.2 MCQ 2017 2M

The expression for an electric field in free space is,

$$\vec{E} = E_0(\hat{x} + \hat{y} + j2\hat{z})e^{-j(\omega t - kx + ky)}$$

where, x, y, z represent the spatial coordinates, t represents time, and ω, k are constants. The electric field

A. does not represent a plane wave.

B. represents a circularly polarized plane wave propagating normal to the z-axis.

C. represents an elliptically polarized plane wave propagating along the x-y plane.

D. represents a linearly polarized plane wave.

Q2.16.1 MCQ 2016 1M

Faraday's law of electromagnetic induction is mathematically described by which one of the following equations?

A. $\nabla \cdot \vec{B} = 0$

B. $\nabla \cdot \vec{D} = \rho_v$

C. $\nabla \times \vec{E} = -\dfrac{\partial \vec{B}}{\partial t}$

D. $\nabla \times \vec{H} = \sigma\vec{E} + \dfrac{\partial \vec{D}}{\partial t}$

Q2.16.2 MCQ 2016 1M

If a right-handed circularly polarized wave is incident normally on a plane perfect conductor, then the reflected wave will be

A. right-handed circularly polarized

B. left-handed circularly polarized

C. elliptically polarized with a tilt angle of 45°

D. horizontally polarized

Q2.16.3 MCQ 2016 1M

Let the electric field vector of a plane electromagnetic wave propagating in a homogenous medium be expressed as,

$$E = \hat{x}E_x e^{-j(\omega t - \beta z)}$$

where the propagation constant β is function of the angular frequency ω. Assume that, $\beta(\omega)$ and E_x are known and are real. From the information available, which one of the following cannot be determined?

A. The type of polarization of the wave.

B. The group velocity of the wave.

C. The phase velocity of the wave.

D. The power flux through the $z = 0$ plane.

Q2.15.1 MCQ 2015 2M

The electric field of a plane wave propagating in a lossless non-magnetic medium is given by the following expression:

$$\vec{E}(z,t) = \hat{a}_x 5\cos(2\pi \times 10^9 t + \beta z)$$
$$+\hat{a}_y 3\cos\left(2\pi \times 10^9 t + \beta z - \frac{\pi}{2}\right)$$

The polarization of the wave is

A. right-handed circular **B.** left-handed elliptical
C. right-handed elliptical **D.** linear

Q2.15.2 MCQ 2015 1M

The electric field of a uniform plane electromagnetic wave is

$$\vec{E} = (\hat{a}_x + j2\hat{a}_y)\exp[j(2\pi \times 10^7 t - 0.2z)]$$

The polarization of the wave is

A. right-handed circular **B.** right-handed elliptical
C. left-handed circular **D.** left-handed elliptical

Q2.15.3 NAT 2015 1M

The electric field intensity of a plane wave travelling in free space is given by the following expression:

$$\vec{E}(x,t) = \hat{a}_y 24\pi\cos(\omega t - k_o x)\ \text{V/m}$$

In this field, consider a square area 10 cm × 10 cm on a plane $x + y = 1$. The total time averaged power (in mW) passing through the square area is _____.

Q2.15.4 NAT 2015 2M

Consider a uniform plane wave with amplitude (E_0) of 10 V/m and 1.1 GHz frequency travelling an air, and incident normally on a dielectric medium with complex relative permittivity (ε_r) and permeability (μ_r) as shown in figure.

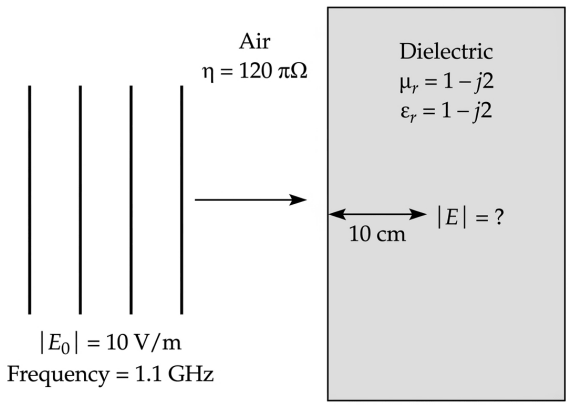


Fig. Q2.15.4

The magnitude of the transmitted electric field component (in V/m) after it has travelled a distance of 10 cm inside the dielectric region is _____.

Q2.15.5 NAT 2015 1M

The electric field component of a plane wave travelling in lossless dielectric medium is given by

$$\vec{E}(z,t) = \hat{a}_y 2\cos\left(10^8 t - \frac{z}{\sqrt{2}}\right)\ \text{V/m}$$

The wavelength (in m) for the wave is _____.

Q2.14.1 NAT 2014 2M

Assume that a plane wave in air with an electric field $\vec{E} = 10\cos(\omega t - 3x - \sqrt{3}z)\hat{a}_y$ V/m is incident on a non-magnetic dielectric slab or relative permittivity 3 which covers the region $z > 0$. The angle of transmission in the dielectric slab is _____ degrees.

Q2.14.2 MCQ 2014 2M

If the electric field of a plane wave is

$$\vec{E}(z,t) = \hat{x}3\cos(\omega t - kz + 30°) - \hat{y}4\sin(\omega t - kz + 45°)$$

mV/m

The polarization state of the plane wave is

A. left elliptical **B.** left circular
C. right elliptical **D.** right circular

Q2.14.3 MCQ 2014 1M

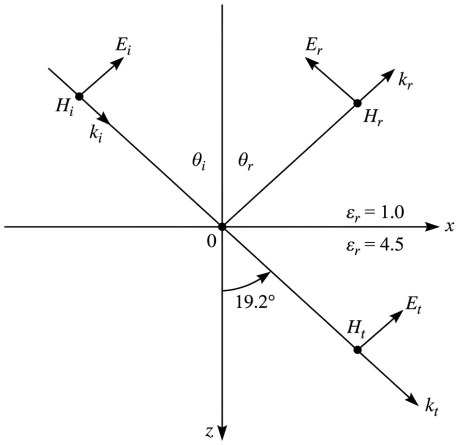
Which one of the following field patterns represents a TEM wave travelling in the positive x-direction?

A. $E = +8\hat{y}, H = -4\hat{z}$ **B.** $E = -2\hat{y}, H = -3\hat{z}$
C. $E = +2\hat{z}, H = +2\hat{y}$ **D.** $E = -3\hat{y}, H = +4\hat{z}$

Common Data Questions

The next 2 questions share the data below.

Statement for Linked Answer Questions (Q2.13.1 and Q2.13.2): A monochromatic plane wave of wavelength $\lambda = 600\,\mu$m is propagating in the direction as shown in the figure below. $\vec{E}_i$, $\vec{E}_r$ and $\vec{E}_t$ denote incident, reflected, and transmitted electric field vectors associated with the wave.



Q2.13.1 MCQ 2013 2M

A monochromatic plane wave of wavelength $\lambda = 600\ \mu m$ is propagating in the direction as shown in the figure below. $\vec{E}_i$, $\vec{E}_r$ and $\vec{E}_t$ denote the incident, reflected and transmitted electric field vectors associated with the wave.

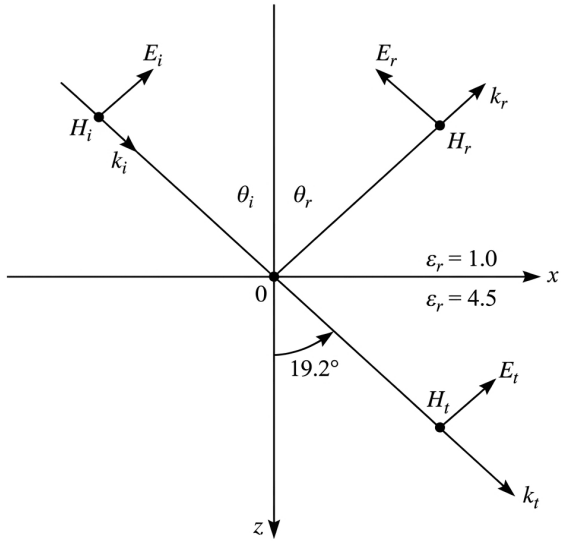


Fig. Q2.13.1

The angle of incidence θ_i and the expression for $\vec{E}_i$ are

A. $60°$ and $\dfrac{E_o}{\sqrt{2}}(\hat{a}_x - \hat{a}_z)e^{-j\frac{\pi \times 10^4 (x+z)}{3\sqrt{2}}}$ V/m

B. $45°$ and $\dfrac{E_o}{\sqrt{2}}(\hat{a}_x + \hat{a}_z)e^{-j\frac{\pi \times 10^4 z}{3}}$ V/m

C. $45°$ and $\dfrac{E_o}{\sqrt{2}}(\hat{a}_x - \hat{a}_z)e^{-j\frac{\pi \times 10^4 (x+z)}{3\sqrt{2}}}$ V/m

D. $60°$ and $\dfrac{E_o}{\sqrt{2}}(\hat{a}_x - \hat{a}_z)e^{-j\frac{\pi \times 10^4 z}{3}}$ V/m

Q2.13.2 MCQ 2013 2M

The expression for $\vec{E}_r$ is

A. $-0.23\dfrac{E_o}{\sqrt{2}}(\hat{a}_x + \hat{a}_z)e^{-j\frac{\pi \times 10^4 (x-z)}{3\sqrt{2}}}$ V/m

B. $-\dfrac{E_o}{\sqrt{2}}(\hat{a}_x + \hat{a}_z)e^{j\frac{\pi \times 10^4 z}{3}}$ V/m

C. $0.44\dfrac{E_o}{\sqrt{2}}(\hat{a}_x + \hat{a}_z)e^{-j\frac{\pi \times 10^4 (x-z)}{3}}$ V/m

D. $\dfrac{E_o}{\sqrt{2}}(\hat{a}_x + \hat{a}_z)e^{-j\frac{\pi \times 10^4 (x+z)}{3}}$ V/m

Q2.12.1 MCQ 2012 1M

A plane wave propagating in air with

$$\vec{E} = (8\hat{a}_x + 6\hat{a}_y + 5\hat{a}_z)e^{j(\omega t + 3x - 4y)}\ \text{V/m}$$

is incident on a perfectly conducting slab positioned at $x \le 0$. The $\vec{E}$ field of the reflected waves is

A. $(-8\hat{a}_x - 6\hat{a}_y - 5\hat{a}_z)e^{j(\omega t + 3x + 4y)}$ V/m

B. $(-8\hat{a}_x + 6\hat{a}_y - 5\hat{a}_z)e^{j(\omega t + 3x + 4y)}$ V/m

C. $(8\hat{a}_x - 6\hat{a}_y - 5\hat{a}_z)e^{j(\omega t - 3x - 4y)}$ V/m

D. $(8\hat{a}_x + 6\hat{a}_y - 5\hat{a}_z)e^{j(\omega t - 3x - 4y)}$ V/m

Q2.12.2 MCQ 2012 1M

The electric field a uniform plane electromagnetic wave in free space, along the positive x-direction, is given by

$$\vec{E} = 10(\hat{a}_y + j\hat{a}_z)e^{-j25x}$$

The frequency and polarization of the wave, respectively are

A. 1.2 GHz and left circular

B. 4 GHz and left circular

C. 1.2 GHz and right circular

D. 4 Hz and right circular

Q2.11.1 MCQ 2011 2M

The electric and magnetic fields for a TEM wave of frequency 14 GHz in a homogeneous medium of relative permittivity ε_r and relative permeability $\mu_r = 1$ are given by

$$\vec{E} = E_p e^{j(\omega t - 280\pi y)} \hat{u}_z \text{ V/m}$$

$$\vec{H} = 3e^{j(\omega t - 280\pi y)} \hat{u}_x \text{ V/m}$$

Assuming the speed of light in free space to be 3×10^8 m/s, intrinsic impedance of free space to be 120π, the relative permittivity ε_r of the medium and the electric field amplitude E_p are

A. $\varepsilon_r = 3, E_p = 120\pi$ **B.** $\varepsilon_r = 3, E_p = 360\pi$
C. $\varepsilon_r = 9, E_p = 360\pi$ **D.** $\varepsilon_r = 9, E_p = 120\pi$

Q2.10.1 MCQ 2010 2M

A plane wave having the electric field component, $\vec{E}_i = 24\cos(3 \times 10^8 t - \beta y)\hat{a}_z$ V/m and travelling in free space is incident normally on a lossless medium with $\mu = \mu_0$ and $\epsilon = 9\epsilon_0$ which occupies the region $y \geq 0$. The reflected magnetic field component is given by

A. $\frac{1}{10\pi}\cos(3 \times 10^8 t + \beta y)\hat{a}_x$ A/m

B. $\frac{1}{20\pi}\cos(3 \times 10^8 t + \beta y)\hat{a}_x$ A/m

C. $-\frac{1}{20\pi}\cos(3 \times 10^8 t + \beta y)\hat{a}_x$ A/m

D. $-\frac{1}{10\pi}\cos(3 \times 10^8 t + \beta y)\hat{a}_x$ A/m

Q2.10.2 MCQ 2010 1M

The electric field component of a time harmonic plane EM wave travelling in a non-magnetic lossless dielectric medium has an amplitude of 1 V/m. If the relative permittivity of the medium is 4, the magnitude of the time average power density vector (in W/m^2) is

A. $\frac{1}{30\pi}$ **B.** $\frac{1}{60\pi}$
C. $\frac{1}{120\pi}$ **D.** $\frac{1}{240\pi}$

Q2.08.1 MCQ 2008 2M

A uniform plane wave in the free space is normally incident on an infinitely thick dielectric slab (dielectric constant $\varepsilon_r = 9$). The magnitude of the reflection coefficient is

A. 0 **B.** 0.3
C. 0.5 **D.** 0.8

Q2.07.1 MCQ 2007 1M

A plane wave of wavelength λ is travelling in a direction making an angle 30° with positive x-axis and 90° with positive y-axis. The $\vec{E}$ field of the plane wave can be represented as (E_0 is a constant)

A. $\vec{E} = \hat{y}E_0 e^{j\left(\omega t - \frac{\sqrt{3}\pi}{\lambda}x - \frac{\pi}{\lambda}z\right)}$

B. $\vec{E} = \hat{y}E_0 e^{j\left(\omega t - \frac{\pi}{\lambda}x - \frac{\sqrt{3}\pi}{\lambda}z\right)}$

C. $\vec{E} = \hat{y}E_0 e^{j\left(\omega t + \frac{\sqrt{3}\pi}{\lambda}x + \frac{\pi}{\lambda}z\right)}$

D. $\vec{E} = \hat{y}E_0 e^{j\left(\omega t - \frac{\pi}{\lambda}x + \frac{\sqrt{3}\pi}{\lambda}z\right)}$

Q2.07.2 MCQ 2007 2M

A Right Circularly Polarized (RCP) plane wave is incident at an angle of 60° to the normal, on an air-dielectric interface. If the reflected wave is linearly polarized, the relative dielectric constant ε_{r2} is

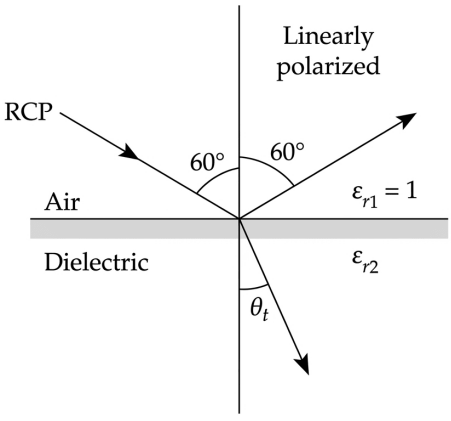


Fig. Q2.07.2

A. $\sqrt{2}$ **B.** $\sqrt{3}$
C. 2 **D.** 3

Q2.07.3 MCQ 2007 2M

The $\vec{H}$ field (in A/m) of a plane wave propagating in free space is given by

$$\vec{H} = \hat{x}\frac{5\sqrt{3}}{\eta_0}\cos(\omega t - \beta z) + \hat{y}\frac{5}{\eta_0}\sin\left(\omega t - \beta z + \frac{\pi}{2}\right)$$

The time average power flow density in watts is

A. $\frac{\eta_0}{100}$ **B.** $\frac{100}{\eta_0}$

C. $50\eta_0^2$ **D.** $\frac{50}{\eta_0}$

Q2.06.1 MCQ 2006 2M

When a plane wave travelling in free-space is incident normally on a medium having $\epsilon_r = 4.0$, the fraction of power transmitted into the medium is given by

A. $\frac{8}{9}$ **B.** $\frac{1}{2}$

C. $\frac{1}{3}$ **D.** $\frac{5}{6}$

Q2.06.2 MCQ 2006 2M

A medium of relative permittivity $\epsilon_{r2} = 2$ forms an interface with free-space. A point source of electromagnetic energy is located in the medium at a depth of 1 meter from the interface. Due to the total internal reflection, the transmitted beam has a circular cross-section over the interface. The area of the beam cross-section at the interface is given by

A. 2π m^2 **B.** π^2 m^2

C. $\frac{\pi}{2}$ m^2 **D.** π m^2

Q2.06.3 MCQ 2006 1M

The electric field on an electromagnetic wave propagating in the positive z-direction is given by

$$E = \hat{a}_x \sin(\omega t - \beta z) + \hat{a}_y \sin\left(\omega t - \beta z + \frac{\pi}{2}\right)$$

The wave is

A. linearly polarized in the z-direction.

B. elliptically polarized.

C. left-hand circularly polarized.

D. right-hand circularly polarized.

Q2.05.1 MCQ 2005 1M

Refractive index of glass is 1.5. Find the wavelength of a beam of light with frequency of 10^{14} Hz in glass. Assume velocity of light is 3×10^8 m/s in vacuum.

A. 3 μm **B.** 3 mm

C. 2 μm **D.** 1 mm

Q2.05.2 MCQ 2005 1M

The magnetic field intensity vector of a plane wave is given by

$$\vec{H}(x,t) = 10\sin(50000t + 0.004x + 30)\hat{a}_y$$

where, $\hat{a}_y$ denotes the unit vector in y-direction. The wave is propagating with a phase velocity

A. 5×10^4 m/s **B.** 3×10^8 m/s

C. 1.25×10^7 m/s **D.** 3×10^6 m/s

Q2.04.1 MCQ 2004 2M

If,

$$\vec{E} = (\hat{a}_x + j\hat{a}_y)\, e^{jkz - j\omega t} \text{ and}$$

$$\vec{H} = \left(\frac{k}{\omega\mu}\right)(\hat{a}_y + j\hat{a}_x)\, e^{jkz - j\omega t}$$

the time averaged pointing vector is

A. null vector **B.** $\left(\frac{k}{\omega\mu}\right)\hat{a}_z$

C. $\left(\frac{2k}{\omega\mu}\right)\hat{a}_z$ **D.** $\left(\frac{k}{2\omega\mu}\right)\hat{a}_z$

Q2.04.2 MCQ 2004 2M

A parallel plate air-filled capacitor has plate area of 10^{-4} m^2 and plate separation of 10^{-3} m. It is connected to a 0.5 V, 3.6 GHz source. The magnitude of the displacement current is ($\varepsilon_0 = 1/36\pi \times 10^{-9}$ F/m)

A. 10 mA **B.** 100 mA

C. 10 A **D.** 1.59 mA

Q2.04.3 MCQ 2004 2M

A plane electromagnetic wave propagating in free space is incident normally on a large slab of lossless, non-magnetic, dielectric material with $\epsilon > \epsilon_0$. Maxima and minima are observed when the electric field is measured in front of the slab. The maximum electric field is found to be 5 times the minimum field. The intrinsic impedance of the medium should be

A. 120 $\pi\Omega$ **B.** 60 $\pi\Omega$

C. 600 $\pi\Omega$ **D.** 24 $\pi\Omega$

Q2.03.1 MCQ 2003 2M

If the electric field intensity associated with a uniform plane electromagnetic wave travelling in a perfect dielectric medium is given by

$$E(z,t) = 10\cos(2\pi \times 10^7 t - 0.1\pi z) \text{ Volt/m}$$

The velocity of the travelling wave is

A. 3.00×10^8 m/sec
B. 2.00×10^8 m/sec
C. 6.28×10^7 m/sec
D. 2.00×10^7 m/sec

Q2.03.2 MCQ 2003 2M

A uniform plane wave travelling in air is incident on the plane boundary between air and another dielectric medium with $\varepsilon_r = 4$. The reflection coefficient for electric field at normal incidence, is

A. zero
B. $0.5\angle 180°$
C. $0.333\angle 0°$
D. $0.333\angle 180°$

Q2.03.3 MCQ 2003 1M

The depth of penetration of electromagnetic wave in a medium having conductivity σ at a frequency of 1 MHz is 25 cm. The depth of penetration at a frequency of 4 MHz will be

A. 6.25 cm
B. 12.50 cm
C. 50.00 cm
D. 100.00 cm

Q2.02.1 MCQ 2002 2M

Distilled water at 25°C is characterized by

$$\sigma = 1.7 \times 10^{-4} \text{ ℧/m and } \epsilon = 78\epsilon_0$$

at a frequency of 3 GHz. Its loss tangent $\tan\delta$ is

A. 1.3×10^{-5}
B. 1.3×10^{-3}
C. $1.7 \times 10^{-4}/78$
D. $1.7 \times 10^{-4}/(78\,\epsilon_0)$

Q2.02.2 MCQ 2002 2M

A plane wave is characterized by

$$\vec{E} = \left(0.5\hat{x} + \hat{y}e^{j\pi/2}\right) e^{j\omega t - jkz}$$

The wave is

A. linearly polarized
B. circularly polarized
C. elliptically polarized
D. unpolarized

Q2.01.1 MCQ 2001 2M

A material has conductivity of 10^{-2} ℧/m and a relative permittivity of 4. The frequency at which the conduction current in the medium is equal to the displacement current is

A. 45 MHz
B. 90 MHz
C. 450 MHz
D. 900 MHz

Q2.01.2 MCQ 2001 1M

If a plane electromagnetic wave satisfies the equation $\frac{\partial^2 E_x}{\partial z^2} = \frac{1}{c^2}\frac{\partial^2 E_x}{\partial t^2}$, the wave propagates in the

A. x-direction
B. z-direction
C. y-direction
D. xy-plane at an angle of 45° between the x and z-directions

Q2.00.1 MCQ 2000 2M

Two coaxial cables 1 and 2 are filled with different dielectric constants ϵ_{r1} and ϵ_{r2} respectively. The ratio of wavelengths in the two cables, (λ_1/λ_2) is

A. $\sqrt{\frac{\epsilon_{r1}}{\epsilon_{r2}}}$
B. $\sqrt{\frac{\epsilon_{r2}}{\epsilon_{r1}}}$
C. $\frac{\epsilon_{r1}}{\epsilon_{r2}}$
D. $\frac{\epsilon_{r2}}{\epsilon_{r1}}$

Q2.00.2 MCQ 2000 2M

A uniform plane wave in air impinges at 45° angle on a lossless dielectric material with dielectric constant ϵ_r. The transmitted wave propagates in a 30° direction with respect to the normal. The value of ϵ_r is

A. 1.5
B. $\sqrt{1.5}$
C. 2
D. $\sqrt{2}$

Q2.00.3 MCQ 2000 1M

A TEM wave is incident normally upon a perfect conductor. The E and H fields at the boundary will be respectively

A. minimum and minimum
B. maximum and maximum
C. minimum and maximum
D. maximum and minimum

Q2.99.1 MCQ 1999 2M

A plane wave propagation through a medium [$\epsilon_r = 8, \mu_r = 2$ and $\sigma = 0$] has its electric field given by

$$\vec{E} = 0.5\sin(10^8 t - \beta z)\ \text{V/m}$$

The wave impedance, (in Ω) is

A. 377 **B.** $198.5\angle 180°$
C. $182.9\angle 14°$ **D.** 188.5

Q2.99.2 MCQ 1999 1M

Identify which one of the following will not satisfy the wave equation?

A. $50e^{j(\omega t - 3z)}$ **B.** $\sin[\omega(10z + 5t)]$
C. $\cos(y^2 + 5t)$ **D.** $\sin(x)\cos(t)$

Q2.98.1 MCQ 1998 1M

The Maxwell's equation, $\nabla \times \vec{H} = \vec{J} + \dfrac{\partial \vec{D}}{\partial t}$ is based on

A. Ampere's law **B.** Gauss's law
C. Faraday's law **D.** Coulomb's law

Q2.98.2 MCQ 1998 1M

A loop is rotating about the y-axis in a magnetic field $\vec{B} = B_0\cos(\omega t + \phi)\hat{a}_x$ T. The voltage in the loop is

A. zero

B. due to rotation only

C. due to transformer action only

D. due to both rotation and transformer action

Q2.98.3 MCQ 1998 1M

The time averaged pointing vector, (in W/m^2), for a wave with $\vec{E} = 24e^{j(\omega t + \beta z)}\hat{a}_y$ V/m in free space is

A. $-\dfrac{2.4}{\pi}\hat{a}_z$ **B.** $\dfrac{2.4}{\pi}\hat{a}_z$
C. $\dfrac{4.8}{\pi}\hat{a}_z$ **D.** $-\dfrac{4.8}{\pi}\hat{a}_z$

Q2.98.4 MCQ 1998 1M

The polarization of a wave with electric field vector $\vec{E} = E_o e^{j(\omega t - \beta z)}(\hat{a}_x + \hat{a}_y)$ is

A. linear **B.** elliptical
C. left hand circular **D.** right hand circular

Q2.98.5 MCQ 1998 1M

The depth of penetration of a wave in a lossy dielectric increases with increasing

A. conductivity **B.** permeability
C. wavelength **D.** permittivity

Q2.98.6 MCQ 1998 1M

The wavelength of a wave with propagation constant $(0.1\pi + j0.2\pi)$ m^{-1} is

A. $\dfrac{2}{\sqrt{0.05}}$ m **B.** 10 m
C. 20 m **D.** 30 m

Q2.98.7 MCQ 1998 1M

The intrinsic impedance of copper at high frequency is

A. purely resistive

B. purely inductive

C. complex with a capacitive component

D. complex with an inductive component

Q2.96.1 MCQ 1996 2M

Some unknown material has a conductivity of 10^6 ℧/m and a permeability of $4\pi \times 10^{-7}$ H/m. The skin depth for the material at 1 GHz is

A. 15.9 μm **B.** 20.9 μm
C. 25.9 μm **D.** 30.9 μm

Q2.96.2 MCQ 1996 2M

A uniform plane wave in air is normally incident on an infinitely thick slab. If the refractive index of the glass slab is 1.5, then the percentage of the incident power that is reflected from the air glass interface is

A. 0% **B.** 4%
C. 20% **D.** 100%

Q2.95.1 MCQ 1995 1M

Copper behaves as a

A. conductor always

B. conductor or dielectric depending on the applied electric field strength

C. conductor or dielectric depending on the frequency

D. conductor or dielectric depending on the electric current density

Q2.95.2 MCQ 1995 1M

The intrinsic impedance of a lossy dielectric medium is given by

A. $\frac{j\omega\mu}{\sigma}$ **B.** $\frac{j\omega\epsilon}{\mu}$

C. $\sqrt{\frac{j\omega\mu}{\sigma + j\omega\epsilon}}$ **D.** $\sqrt{\frac{\mu}{\epsilon}}$

Q2.94.1 MCQ 1994 1M

A plane electromagnetic wave travelling along the $+z$-direction, has its electric field given by $E_x = 2\cos(\omega t)$ and $E_y = 2\cos(\omega t + 90°)$ The wave is

A. linearly polarized **B.** right circularly polarized
C. left circularly polarized **D.** elliptically polarized

Q2.93.1 MSQ 1993 2M

A plane wave is incident normally on a perfect conductor as shown in figure. Here E_x^i, H_y^i and $\vec{p}^i$ are electric field, magnetic field and pointing vector, respectively, for the incident wave. The reflected waver should have

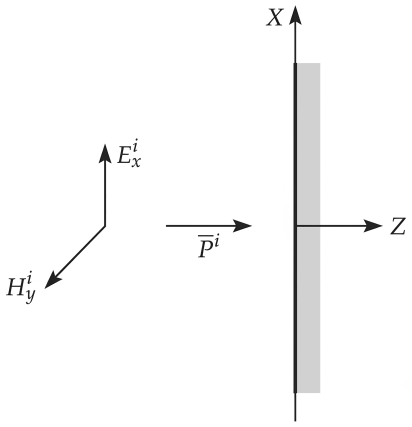


Fig. Q2.93.1

A. $E_x^r = -E_x^i$ **B.** $H_y^r = -H_y^i$
C. $\vec{p}^r = -\vec{p}^i$ **D.** $E_x^r = E_x^i$

Q2.93.2 MCQ 1993 2M

A material is described by the following electrical parameters at a frequency of 10 GHz, $\sigma = 10^6$ ℧/m, $\mu = \mu_0$ and $\epsilon/\epsilon_0 = 10$. The material at this frequency is considered to be $\left(\epsilon_0 = \frac{1}{36\pi} \times 10^{-9}\ \text{F/m}\right)$.

A. a good conductor

B. a good dielectric

C. neither a good conductor, nor a good dielectric

D. a good magnetic material

Q2.91.1 MSQ 1991 2M

The electric field component of a uniform plane electromagnetic wave propagation in the Y-direction in a lossless medium will satisfy the equation

☐ **A.** $\frac{\partial^2 E_y}{\partial y^2} = \mu\epsilon\frac{\partial^2 E_y}{\partial t^2}$

☐ **B.** $\frac{\partial^2 E_y}{\partial x^2} = \mu\epsilon\frac{\partial^2 E_y}{\partial t^2}$

☐ **C.** $\frac{\partial^2 E_x}{\partial y^2} = \mu\epsilon\frac{\partial^2 E_x}{\partial t^2}$

☐ **D.** $\frac{\sqrt{E_x^2 + E_z^2}}{\sqrt{H_x^2 + H_z^2}} = \sqrt{\frac{\mu}{\epsilon}}$

Q2.90.1 MCQ 1990 2M

The incoming solar radiation at a place on the surface of the earth is 1.2 kW/m^2. The amplitude of the electric field corresponding to this incident power is nearly equal to

A. 80 mV/m **B.** 2.5 V/m
C. 30 V/m **D.** 950 V/m

Q2.89.1 MCQ 1989 2M

The skin depth of copper at a frequency of 3 GHz is 1 micro (10^{-6} meter). At 12 GHz, for a non magnetic conductor whose conductivity is 1/9 times that of copper, the skin depth would be

A. $\sqrt{9\times 4}$ microns **B.** $\sqrt{\frac{9}{4}}$ microns

C. $\sqrt{\frac{4}{9}}$ microns **D.** $\frac{1}{\sqrt{9\times 4}}$ microns

Q2.88.1 MCQ 1988 2M

In a good conductor the phase relation between the tangential components of electric field E_t and the magnetic field H_t is as follows:

A. E_t and H_t are in phase.

B. E_t and H_t are out of phase.

C. H_t leads E_t by 90°.

D. E_t leads H_t by 45°.

Q2.87.1 MCQ 1987 2M

For an electromagnetic wave incident from one medium to a second medium, total reflection takes place when

A. the angle of incidence is equal to the Brewster angle with E field perpendicular to the plane of incidence.

B. the angle of incidence is equal to the Brewster angle with E field parallel to the plane of incidence.

C. the angle of incidence is equal to the critical angle with the wave moving from the denser medium to a rarer medium.

D. the angle of incidence is equal to the critical angle with the wave moving from a rarer medium to a rarer medium.

Answer Key

Q2.26.1 A	Q2.26.2 A,B	Q2.24.1 D
Q2.23.1 A,B,D	Q2.21.1 D	Q2.20.1 1
Q2.18.1 B	Q2.18.2 2	Q2.17.1 4.33
Q2.17.2 C	Q2.16.1 C	Q2.16.2 B
Q2.16.3 D	Q2.15.1 B	Q2.15.2 D
Q2.15.3 53.0 to 53.6	Q2.15.4 0.1	Q2.15.5 8.85 to 8.90
Q2.14.1 30	Q2.14.2 A	Q2.14.3 B
Q2.13.1 C	Q2.13.2 A	Q2.12.1 C
Q2.12.2 A	Q2.11.1 D	Q2.10.1 A
Q2.10.2 C	Q2.08.1 C	Q2.07.1 A
Q2.07.2 D	Q2.07.3 D	Q2.06.1 A
Q2.06.2 D	Q2.06.3 C	Q2.05.1 C
Q2.05.2 C	Q2.04.1 A	Q2.04.2 A
Q2.04.3 D	Q2.03.1 B	Q2.03.2 D
Q2.03.3 B	Q2.02.1 A	Q2.02.2 C
Q2.01.1 A	Q2.01.2 B	Q2.00.1 B
Q2.00.2 C	Q2.00.3 C	Q2.99.1 D
Q2.99.2 C	Q2.98.1 A	Q2.98.2 D
Q2.98.3 A	Q2.98.4 A	Q2.98.5 D
Q2.98.6 B	Q2.98.7 D	Q2.96.1 A
Q2.96.2 B	Q2.95.1 A	Q2.95.2 C
Q2.94.1 C	Q2.93.1 A,C	Q2.93.2 A
Q2.91.1 C,D	Q2.90.1 D	Q2.89.1 B
Q2.88.1 D	Q2.87.1 C	

UNIT

III

Transmission Lines & Smith Chart

Topics

1. Types of Transmission Lines
2. Impedance Calculations
3. Standing Wave Patterns
4. DC Analysis
5. Power Transfer
6. Scattering Parameters
7. Smith Chart

PrepFusion

Unit Snapshot*

Stream: EC	**Questions:** 73	**MCQ:** 53	**MSQ:** 3	**NAT:** 17
1M: 28	**2M:** 45	**Descriptive:** 0	**Avg Weightage** ~3M	**Range:** 1M(2013)–8M(2023)

**See preface for how Avg Weightage and Range are calculated.*

Q3.26.1 MCQ 2026 2M

A complex load (in Ω) is represented as $\Gamma_L = 0.5\angle 30°$ on the Smith chart. A co-axial cable with a characteristic impedance of 50 Ω is connected to the load. The new input impedance of the load now moves to a diametrically opposite point on the same Γ circle on the Smith chart. Which option is the nearest input impedance of the cable connected load (in Ω)?

A. 20.7 - j5.1 **B.** 17.7 - j11.8
C. 97.5 - j65.0 **D.** 97.5 + j65.0

Q3.26.2 NAT 2026 1M

For a lossless passive two-port network, $|S_{11}|$ and $|S_{21}|$ intersect at −3 dB. For a lossy passive two-port network, $|S_{11}|$ and $|S_{21}|$ intersect at −4 dB. The percentage of power dissipated in the lossy network at the intersection frequency is _____ (rounded off to two decimal places)

Q3.25.1 NAT 2025 2M

A 50 Ω lossless transmission line is terminated with a load Z_L of $(50 - j75)$ Ω. If the average incident power on the line is 10 mW, then the average power delivered to the load (in mW, rounded off to one decimal place) is _____.

Q3.24.1 NAT 2024 2M

A lossless transmission line with characteristic impedance $Z_0 = 50$ Ω is terminated with an unknown load. The magnitude of the reflection coefficient is $|\Gamma| = 0.6$. As one moves towards the generator from the load, the maximum value of the input impedance magnitude looking towards the load (in Ω) is _____.

Q3.24.2 MCQ 2024 2M

Consider a lossless transmission line terminated with a short circuit as shown in the figure below. As one moves towards the generator from the load, the normalized impedances z_{inA}, z_{inB}, z_{inC} and z_{inD} (indicated in the figure) are _____.

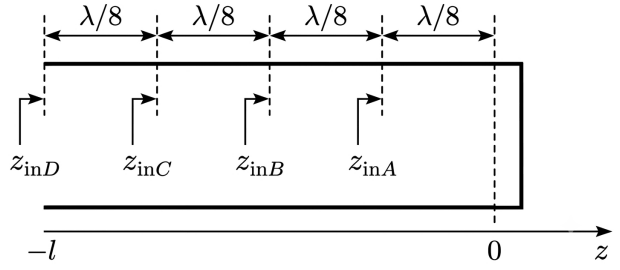


Fig. Q3.24.2

A. $z_{inA} = +0.4j\ \Omega$, $z_{inB} = \infty$, $z_{inC} = -0.4j\ \Omega$, $z_{inD} = 0$

B. $z_{inA} = -1j\ \Omega$, $z_{inB} = 0$, $z_{inC} = +1j\ \Omega$, $z_{inD} = \infty$

C. $z_{inA} = \infty$, $z_{inB} = +0.4j\ \Omega$, $z_{inC} = 0$, $z_{inD} = +0.4j\ \Omega$

D. $z_{inA} = +1j\ \Omega$, $z_{inB} = \infty$, $z_{inC} = -1j\ \Omega$, $z_{inD} = 0$

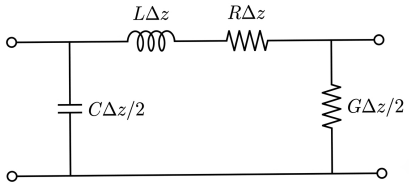
Q3.23.1 NAT 2023 2M

A transparent dielectric coating is applied to glass ($\epsilon_r = 4, \mu_r = 1$) to eliminate the reflection of red light ($\lambda_0 = 0.75\mu$m). The minimum thickness of the dielectric coating, in μm, that can be used is _____ (rounded off to two decimal places).

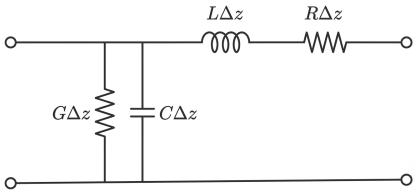
Q3.23.2 MSQ 2023 2M

The following circuit(s) representing a lumped element equivalent of an infinitesimal section of a transmission line is/are

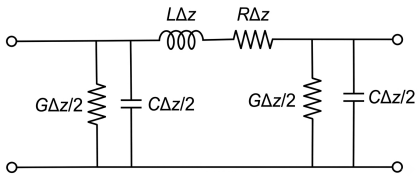
☐ **A.**



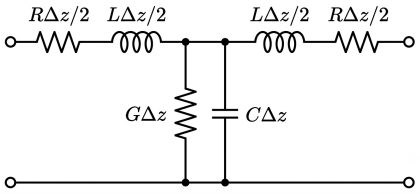
☐ **B.**



☐ **C.**



☐ **D.**



Q3.23.3 MSQ 2023 2M

The standing wave ratio on a 50 Ω lossless transmission line terminated in an unknown load impedance is found to be 2.0. The distance between successive voltage minima is 30 cm and the first minimum is located at 10 cm from the load. Z_L can be replaced by an equivalent length l_m and terminating resistance R_m of the same line. The value of R_m and l_m, respectively, are

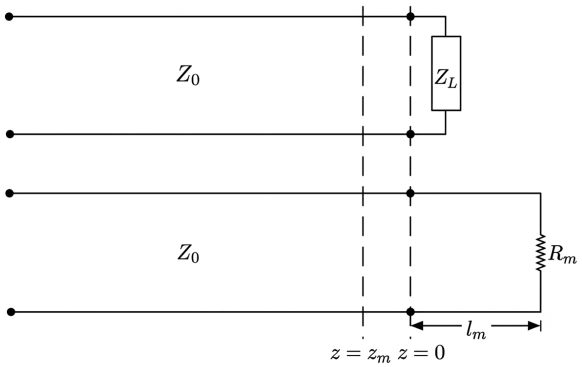


Fig. Q3.23.3

- **A.** $R_m = 100\ \Omega, l_m = 20$ cm
- **B.** $R_m = 25\ \Omega, l_m = 20$ cm
- **C.** $R_m = 100\ \Omega, l_m = 5$ cm
- **D.** $R_m = 25\ \Omega, l_m = 5$ cm

Q3.23.4 MCQ 2023 2M

The S-parameters of a two port network are given as

$$[S] = \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix}$$

with reference to Z_0. Two lossless transmission line sections of electrical lengths $\theta_1 = \beta l_1$ and $\theta_2 = \beta l_2$ are added to the input and output ports for measurement purposes, respectively. The S-parameters $[S']$ of the resultant two port network are

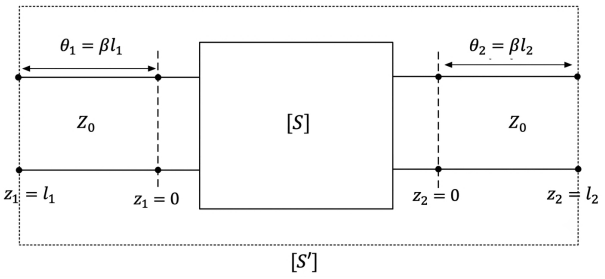


Fig. Q3.23.4

- **A.** $\begin{bmatrix} S_{11}e^{-j2\theta_1} & S_{12}e^{-j(\theta_1+\theta_2)} \\ S_{21}e^{-j(\theta_1+\theta_2)} & S_{22}e^{-j2\theta_2} \end{bmatrix}$
- **B.** $\begin{bmatrix} S_{11}e^{j2\theta_1} & S_{12}e^{-j(\theta_1+\theta_2)} \\ S_{21}e^{-j(\theta_1+\theta_2)} & S_{22}e^{j2\theta_2} \end{bmatrix}$
- **C.** $\begin{bmatrix} S_{11}e^{j2\theta_1} & S_{12}e^{j(\theta_1+\theta_2)} \\ S_{21}e^{j(\theta_1+\theta_2)} & S_{22}e^{j2\theta_2} \end{bmatrix}$
- **D.** $\begin{bmatrix} S_{11}e^{-j2\theta_1} & S_{12}e^{j(\theta_1+\theta_2)} \\ S_{21}e^{j(\theta_1+\theta_2)} & S_{22}e^{-j2\theta_2} \end{bmatrix}$

Q3.21.1 MCQ 2021 2M

The impedance matching network shown in the figure is to match a lossless line having characteristic impedance $Z_0 = 50\ \Omega$ with a load impedance Z_L. A quarter wave line having a characteristic impedance $Z_1 = 75\ \Omega$ is connected to Z_L. Two stubs having characteristic impedance of 75 Ω each are connected to this quarter wave line. One is a short-circuited (S.C.) stub of length of 0.25λ connected across PQ and the other one is an open-circuit (O.C.) stub of length 0.5λ connected across RS.

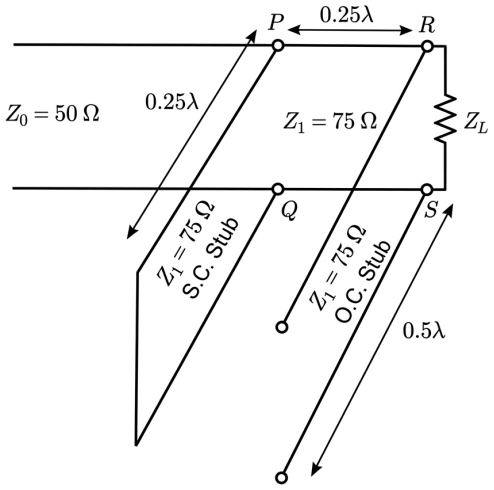


Fig. Q3.21.1

The impedance matching is achieved when the real part of Z_L is

A. 50.0 Ω

B. 33.3 Ω

C. 75.0 Ω

D. 112.5 Ω

Q3.20.1 MCQ 2020 1M ○○○

The impedances $Z = jX$ for all X in the range $(-\infty, \infty)$, map to the Smith chart as,

A. a circle of radius 0.5 with center at $(0.5, 0)$.

B. a point at the center of the chart.

C. a line passing through the center of the chart.

D. a circle of radius 1 with center at $(0, 0)$.

Q3.20.2 NAT 2020 1M ○○○

A transmission line of length $3\lambda/4$ and having a characteristic impedance of 50 Ω is terminated with a load of 400 Ω. The impedance (rounded off to two decimal places) seen at the input end of the transmission line is _____ Ω.

Q3.18.1 NAT 2018 1M ○○○

A lossy transmission line has resistance per unit length $R = 0.05$ Ω/m. The line is distortionless and has characteristic impedance of 50 Ω. The attenuation constant (in Np/m correct to three decimal places) of the line is _____.

Q3.18.2 MCQ 2018 1M ○○○

The points P, Q and R shows on the Smith chart (normalized impedance chart) in the following figure represent:

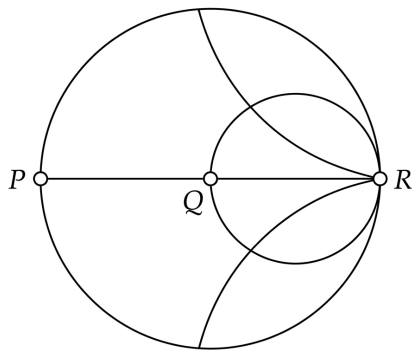


Fig. Q3.18.2

A. P : Open-circuit, Q : Short-circuit, R : Matched load

B. P : Open-circuit, Q : Matched load, R : Short-circuit

C. P : Short circuit, Q : Matched load, R : Open-circuit

D. P : Short circuit, Q : Open-circuit, R : Matched load

Q3.17.1 NAT 2017 2M ○○○

An optical fibre is kept along the $\hat{z}$-direction. The refractive indices for the electric fields along $\hat{x}$ and $\hat{y}$-directions in the fibre are $n_x = 1.5000$ and $n_y = 1.5001$, respectively ($n_x \neq n_y$ due to the imperfection in the fibre cross-section). The free space wavelength of a light wave propagating in the fibre is 1.5 μm. If the light wave is circularly polarized at the input of the fibre, the minimum propagation distance after which is becomes linearly polarized (in centimeters), is _____.

Q3.17.2 NAT 2017 1M ○○○

A two-wire transmission line terminates in a television set. The VSWR measured on the line is 5.8. The percentage of power that is reflected from the television set is _____.

Q3.17.3 NAT 2017 1M ○○○

The voltage of an electromagnetic wave propagating in a coaxial cable with uniform characteristic impedance is $V(l) = e^{-\gamma l + j\omega t}$ Volts, where l is the distance along the length of the cable in meters, $\gamma = (0.1 + j40)$ m^{-1} is the complex propagation constant, and $\omega = 2\pi \times 10^6$ rad/s is the angular frequency. The absolute value of the attenuation in the cable (in dB/meter) is _____.

Q3.16.1 NAT 2016 1M ○○○

Light from free space is incident at an angle θ_i to the normal of the face of a step-index large core optical fibre. The core and cladding refractive indices are $n_1 = 1.5$ and $n_2 = 1.4$, respectively.

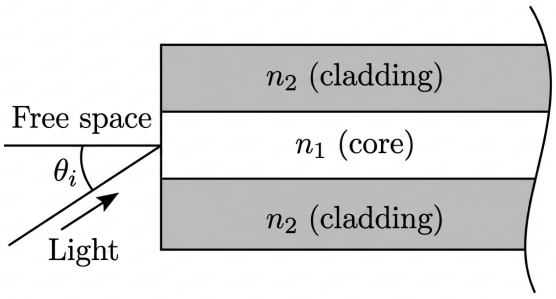


Fig. Q3.16.1

The maximum value of θ_i (in degrees) for which the incident light will be guided in the core of the fibre is _____.

Q3.16.2 NAT 2016 2M

A microwave circuit consisting of lossless transmission lines T_1 and T_2 is shown in the figure. The plot shows the magnitude of the input reflection coefficient $|\Gamma|$ as a function f. The phase velocity of the signal in the transmission lines is 2×10^8 m/s.

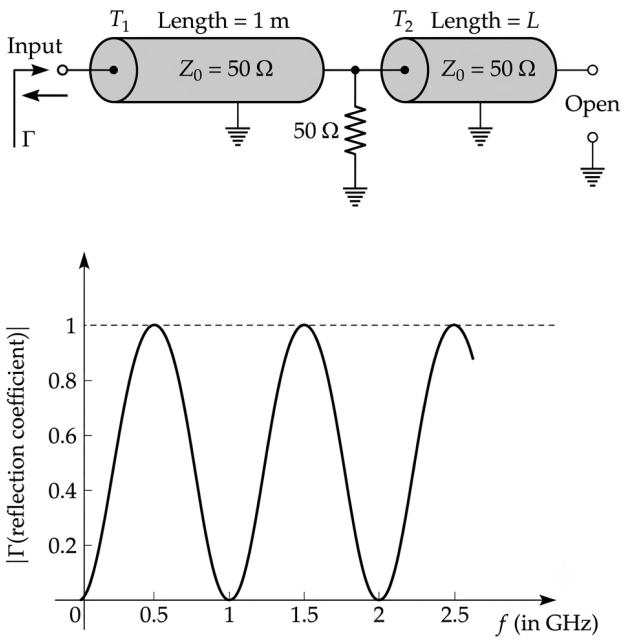


Fig. Q3.16.2

The length L (in meters) of T_2 is _____.

Q3.16.3 MCQ 2016 2M

A lossless microstrip transmission line consists of a trace of width w. It is drawn over a practically infinite ground plane and is separated by a dielectric slab of thickness t and relative permittivity $\varepsilon_r > 1$. The inductance per unit length and the characteristic impedance of this line are L and Z_0, respectively.

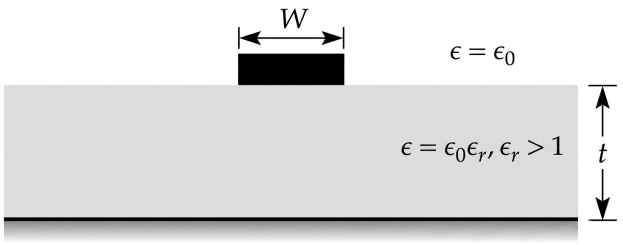


Fig. Q3.16.3

Which one of the following inequalities is always satisfied?

A. $Z_0 > \sqrt{\dfrac{Lt}{\varepsilon_0 \varepsilon_r w}}$ **B.** $Z_0 < \sqrt{\dfrac{Lt}{\varepsilon_0 \varepsilon_r w}}$

C. $Z_0 > \sqrt{\dfrac{Lw}{\varepsilon_0 \varepsilon_r t}}$ **D.** $Z_0 < \sqrt{\dfrac{Lw}{\varepsilon_0 \varepsilon_r t}}$

Q3.16.4 MCQ 2016 1M

The propagation constant of a lossy transmission line is $(2 + j5)$ m^{-1} and its characteristic impedance is $(50 + j0)\,\Omega$ at $\omega = 10^6$ rad s^{-1}. The value of the line constant L, C, R, G are respectively

A. $L = 200\,\mu$H/m, $C = 0.1\,\mu$F/m, $R = 50\,\Omega$/m, $G = 0.02$ S/m

B. $L = 250\,\mu$H/m, $C = 0.1\,\mu$F/m, $R = 100\,\Omega$/m, $G = 0.04$ S/m

C. $L = 200\,\mu$H/m, $C = 0.2\,\mu$F/m, $R = 100\,\Omega$/m, $G = 0.02$ S/m

D. $L = 250\,\mu$H/m, $C = 0.2\,\mu$F/m, $R = 50\,\Omega$/m, $G = 0.04$ S/m

Q3.15.1 NAT 2015 2M

A 200 m long transmission line having parameters shown in the figure is terminated into a load R_L. The line is connected to a 400 V source having source R_s through a switch which is closed at $t = 0$. The transient response of the circuit at the input of the line ($z = 0$) is also drawn in the figure.

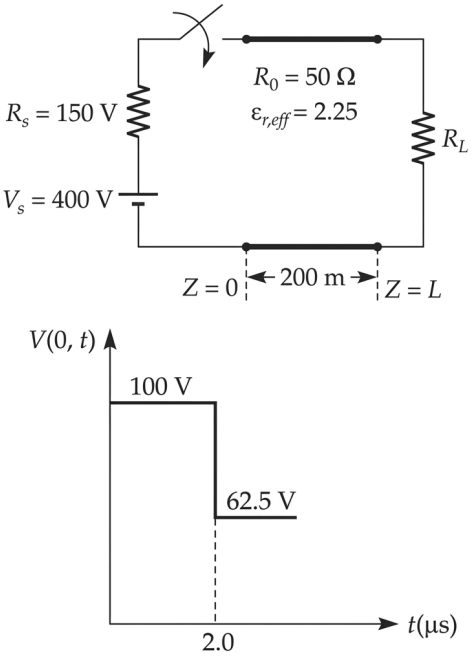


Fig. Q3.15.1

The value of R_L (in Ω) is _____.

Q3.15.2 NAT | 2015 | 2M

A coaxial capacitor of inner radius 1 mm and outer radius 5 mm has a capacitance a per unit length of 172 pF/m. If the ratio of outer radius to inner is doubled, the capacitance per unit length (in pF/m) is _____.

Q3.15.3 MCQ | 2015 | 2M

Consider the 3 m long lossless air-filled transmission line shown in the figure. It has a characteristic impedance of $120\pi\ \Omega$, is terminated by a short-circuit, and is excited with a frequency of 37.5 MHz. What is the nature of the input impedance (Z_{in})?

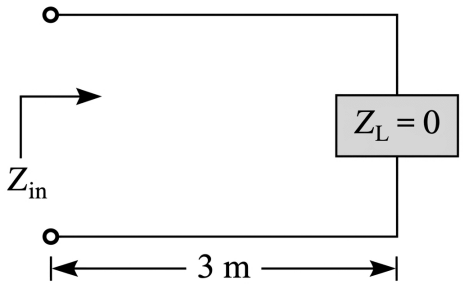


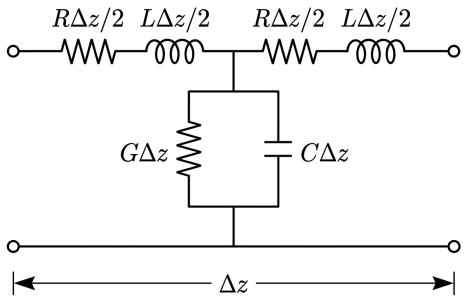
Fig. Q3.15.3

A. Open
B. Short
C. Inductive
D. Capacitive

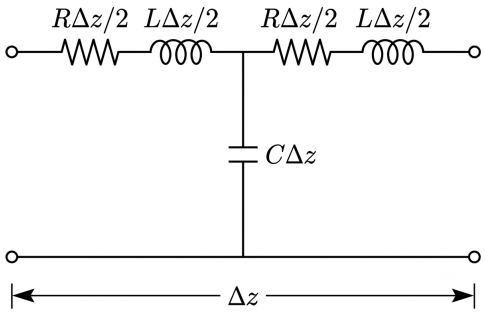
Q3.15.4 MCQ | 2015 | 1M

A co-axial cable is made of two brass conductors. The spacing between the conductors is filled with Teflon ($\epsilon_r = 2.1, \tan\delta = 0$). Which of the following circuits can represent the lumped element model of a small piece of this cable having length Δz?

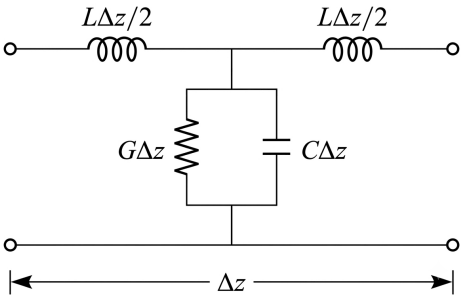
A.



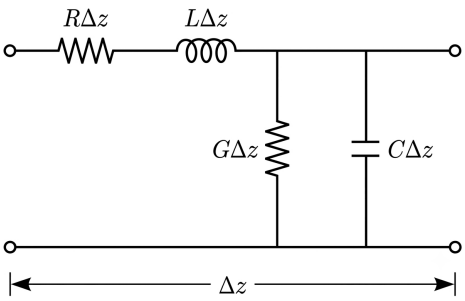
B.



C.



D.



Q3.14.1 MCQ 2014 1M

A two-port network has scattering parameters given by

$$[S] = \begin{bmatrix} s_{11} & s_{12} \\ s_{21} & s_{22} \end{bmatrix}$$

If the port-2 of the two-port is short circuited, the s_{11} parameter for the resultant one-port network is

A. $\dfrac{s_{11} - s_{11}s_{22} + s_{12}s_{21}}{1 + s_{22}}$

B. $\dfrac{s_{11} + s_{11}s_{22} - s_{12}s_{21}}{1 + s_{22}}$

C. $\dfrac{s_{11} + s_{11}s_{22} + s_{12}s_{21}}{1 - s_{22}}$

D. $\dfrac{s_{11} - s_{11}s_{22} + s_{12}s_{21}}{1 - s_{22}}$

Q3.14.2 MCQ 2014 1M

In the following figure, the transmitter T_x sends a wideband modulated RF signal via a coaxial cable to the receiver R_x. The output impedance Z_T of T_x, the characteristic impedance Z_0 of the cable and the input impedance Z_R of R_x are all real.

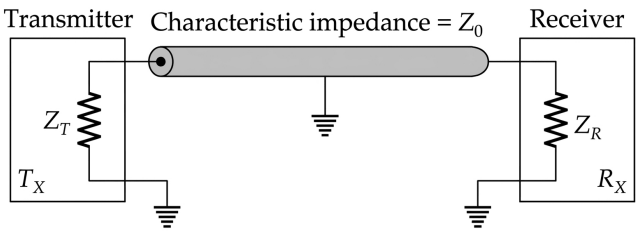


Fig. Q3.14.2

Which one of the following statements is true about the distortion of the received signal due to impedance mismatch?

A. The signal gets distorted if $Z_R \neq Z_0$ irrespective of the value of Z_T.

B. The signal gets distorted if $Z_T \neq Z_0$, irrespective of the value of Z_R.

C. Signal distortion implies impedance mismatch at both ends, $Z_T \neq Z_0$ and $Z_R \neq Z_0$.

D. Impedance mismatches do not result in signal distortion but reduce power transfer efficiency.

Q3.14.3 NAT 2014 2M

In the transmission line shown, the impedance Z_{in} (in Ω) between node A and the ground is _____.

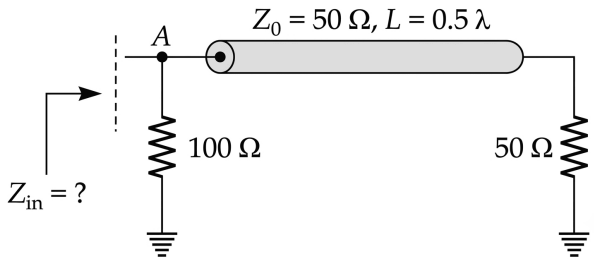


Fig. Q3.14.3

Q3.14.4 NAT 2014 1M

To maximize power transfer, a lossless transmission line is to be matched to a resistive load impedance via a $\lambda/4$ transformer as shown.

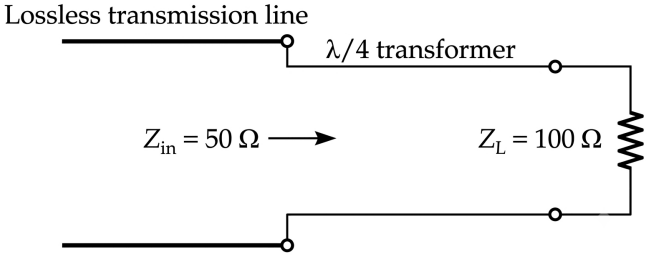


Fig. Q3.14.4

The characteristic impedance (in Ω) of the $\lambda/4$ transformer is _____.

Q3.14.5 NAT 2014 2M

The input impedance of $\lambda/8$ section of a lossless transmission line of characteristic impedance 50 Ω is found to be real when the other end is terminated by a load $Z_L = (R + jX)$ Ω. If X is 30 Ω, the value of R (in Ω) is _____.

Q3.14.6 MCQ 2014 2M

For a parallel plate transmission line, let v be the speed of propagation and z be the characteristic impedance. Neglecting fringe effects, a reduction of the spacing between the plates by a factor of two results in

A. having of v and no change in z.

B. no changes in v and having of z.

C. no change in both v and z.

D. halving of both v and z.

Q3.13.1 MCQ 2013 1M

The return loss of a device is found to be 20 dB. The voltage standing wave ratio (VSWR) and magnitude of reflection coefficient are respectively

A. 1.22 and 0.1 **B.** 0.81 and 0.1
C. -1.22 and 0.1 **D.** 2.44 and 0.2

Q3.12.1 MCQ 2012 2M

A transmission line with a characteristic impedance of 100 Ω is used to match a 50 Ω section to a 200 Ω section. If the matching is to be done both at 429 MHz and 1 GHz, the length of the transmission line can be approximately

A. 82.5 cm **B.** 1.05 m
C. 1.58 m **D.** 1.75 m

Q3.12.2 MCQ 2012 1M

A coaxial cable with an inner diameter of 1 mm and outer diameter of 2.4 mm is filled with a dielectric of relative permittivity 10.89. Given,

$$\mu_0 = 4\pi \times 10^{-7} \text{ H/m}, \ \varepsilon_0 = \frac{10^{-9}}{36\pi} \text{ F/m},$$

the characteristic impedance of the cable as

A. 330 Ω **B.** 100 Ω
C. 143.3 Ω **D.** 43.4 Ω

Q3.11.1 MCQ 2011 2M

A transmission line of characteristic impedance 50 Ω is terminated in a load impedance Z_L. The VSWR of the line is measured as 5 and the first of the voltage maxima in the line is observed at a distance of $\lambda/4$ from the load. The value of Z_L is

A. 10 Ω **B.** 250 Ω
C. (19.23 + j46.15) Ω **D.** (19.23 - j46.15) Ω

Q3.11.2 MCQ 2011 1M

A transmission line of characteristic impedance 50 Ω is terminated by a 50 Ω load. When excited by a sinusoidal voltage source at 10 GHz, the phase difference between two points spaced 2 mm apart on the line is found to be $\pi/4$ radians. The phase velocity of the wave along the line is

A. 0.8×10^8 m/s **B.** 1.2×10^8 m/s
C. 1.6×10^8 m/s **D.** 3×10^8 m/s

Q3.10.1 MCQ 2010 1M

If the scattering matrix $[S]$ of a two-port network is

$$[S] = \begin{bmatrix} 0.2\angle 0^\circ & 0.9\angle 90^\circ \\ 0.9\angle 90^\circ & 0.1\angle 90^\circ \end{bmatrix}$$

then the network is

A. lossless and reciprocal

B. lossless but not reciprocal

C. not lossless but reciprocal

D. neither lossless nor reciprocal

Q3.10.2 MCQ 2010 1M

In the circuit shown, all the transmission line sections are lossless. The Voltage Standing Wave Ratio (VSWR) on the 60 Ω line is

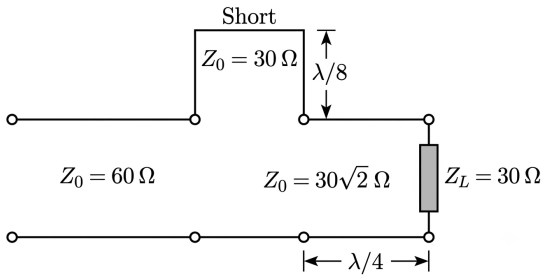


Fig. Q3.10.2

A. 1.00 **B.** 1.64 **C.** 2.50 **D.** 3.00

Q3.10.3 MCQ 2010 1M

A transmission line has a characteristic impedance of 50 Ω and a resistance of 0.1 Ω/m. If the line is distortionless, the attenuation constant (in Np/m) is

A. 500 **B.** 5
C. 0.014 **D.** 0.002

Q3.09.1 MCQ 2009 2M

A transmission line terminates in two branches, each of length $\lambda/4$, as shown. The branches are terminated by 50 Ω loads. The lines are lossless and have the characteristic impedances shown. Determine the impedance Z_i as seen by the source.

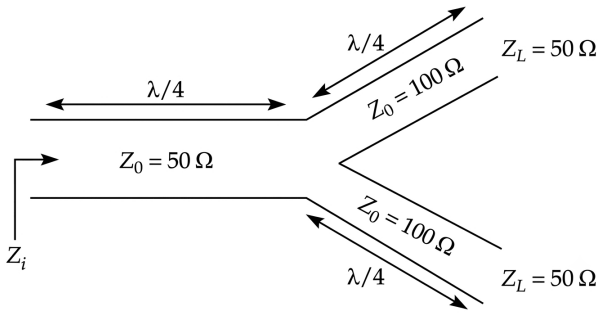


Fig. Q3.09.1

A. 200 Ω **B.** 100 Ω
C. 50 Ω **D.** 25 Ω

Q3.08.1 MCQ 2008 2M

One end of a lossless transmission line having the characteristic impedance of 75 Ω and length of 1 cm is short-circuited. At 3 GHz, the input impedance at the other end of the transmission line is

A. 0 **B.** resistive
C. capacitive **D.** inductive

Q3.08.2 MCQ 2008 2M

In the design of a single mode step index optical fiber close to upper cut-off, the single-mode operation is not preserved if

A. radius as well as operating wavelength are halved.

B. radius as well as operating wavelength are doubled.

C. radius is halved and operating wavelength is doubled.

D. radius is doubled and operating wavelength is halved.

Q3.07.1 MCQ 2007 2M

A load of 50 Ω is connected in shunt in a 2-wire transmission line of $Z_0 = 50\ \Omega$ as shown in the figure. The 2-port scattering parameter matrix (S-matrix) of the shunt element is

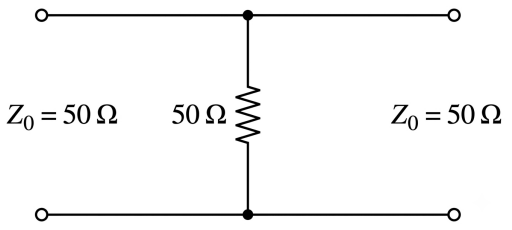


Fig. Q3.07.1

A. $\begin{bmatrix} -\frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{bmatrix}$ **B.** $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

C. $\begin{bmatrix} -\frac{1}{3} & \frac{2}{3} \\ \frac{2}{3} & -\frac{1}{3} \end{bmatrix}$ **D.** $\begin{bmatrix} \frac{1}{4} & -\frac{3}{4} \\ -\frac{3}{4} & \frac{1}{4} \end{bmatrix}$

Q3.07.2 MCQ 2007 2M

The parallel branches of a 2-wire transmission line are terminated in 100 Ω and 200 Ω resistors as shown in the figure. The characteristic impedance of the line is $Z_0 = 50\ \Omega$ and each section has a length of $\lambda/4$. The voltage reflection coefficient Γ at the input is

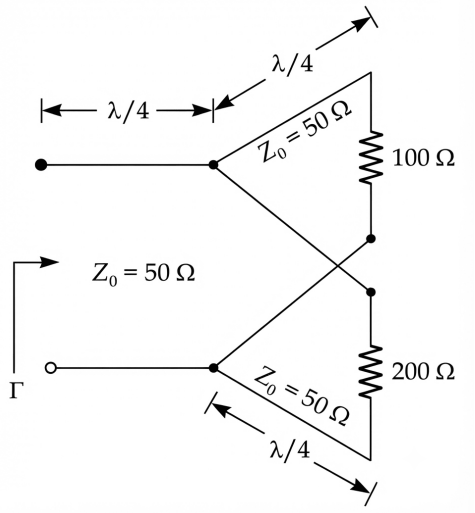


Fig. Q3.07.2

A. $-j\frac{7}{5}$ **B.** $\frac{-5}{7}$ **C.** $j\frac{5}{7}$ **D.** $\frac{5}{7}$

Common Data Questions

The next 2 questions share the data below.

Common Data for Questions(Q3.06.1 and Q3.06.2) :
A 30 Volts a battery with zero source resistance is connected to a co-axial line of characteristic impedance of 50 Ω at $t = 0$ seconds and terminated in an unknown resistive load. The line length is such that it takes 400 μs for an electromagnetic wave to travel from source end to load end and vice-versa. At $t = 400\ \mu$s, the voltage at the load end is found to be 40 Volts.

Q3.06.1 MCQ 2006 2M

The load resistance is

A. 25 Ω **B.** 50 Ω
C. 75 Ω **D.** 100 Ω

Q3.06.2 MCQ 2006 2M

The steady-state current through the load resistance is

A. 1.2 Amps. **B.** 0.3 Amps.
C. 0.6 Amps. **D.** 0.4 Amps.

Q3.05.1 MCQ 2005 2M

Many circles are drawn in a Smith chart used for transmission line calculations. The circles shown in the figure represent

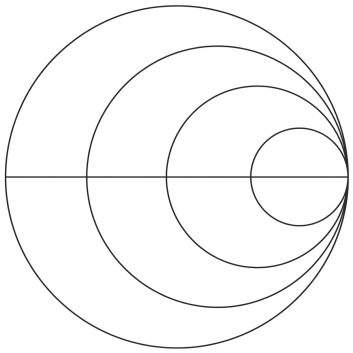


Fig. Q3.05.1

A. unit circles

B. constant resistance circles

C. constant reactance circles

D. constant reflection coefficient circles

Common Data Questions

The next 2 questions share the data below.

Common Data for Questions (Q3.05.2 and Q3.05.3): Voltage standing wave pattern in a lossless transmission line with characteristic impedance 50 Ω and a resistive load is shown in the figure.

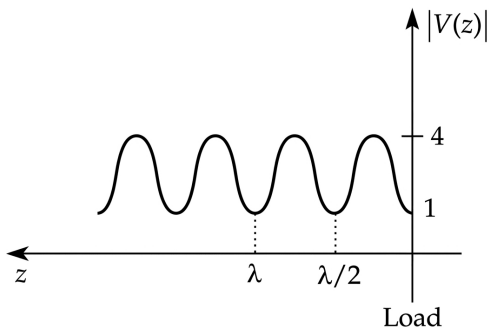


Fig. Q3.05.1

Q3.05.2 MCQ 2005 2M

The value of the load resistance is

A. 50 Ω **B.** 200 Ω
C. 12.5 Ω **D.** 0 Ω

Q3.05.3 MCQ 2005 2M

The reflection coefficient is given by

A. -0.6 **B.** -1
C. 0.6 **D.** 0

Q3.05.4 MCQ 2005 2M

Characteristic impedance of a transmission line is 50 Ω. Input impedance of the open-circuited line is $Z_{OC} = 100 + j150$ Ω. When the transmission line is short-circuited, then value of the input impedance will be

A. 50 Ω **B.** $100 + j150$ Ω
C. $7.69 + j11.54$ Ω **D.** $7.69 - j11.54$ Ω

Q3.04.1 MCQ 2004 2M

A lossless transmission line is terminated in a load which reflects a part of the incident power. The measured VSWR is 2. The percentage of the power that is reflected back is

A. 57.73 **B.** 33.33
C. 0.11 **D.** 11.11

Q3.04.2 MCQ 2004 2M

Consider an impedance $Z = R + jX$ marked with point P in an impedance Smith chart as shown in the figure. The movement from point P along a constant resistance circle in the clockwise direction by an angle 45° is equivalent to

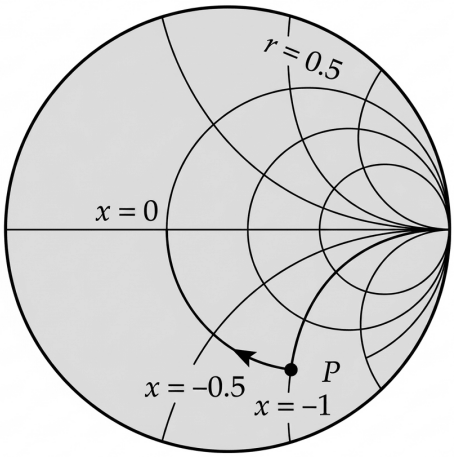


Fig. Q3.04.2

A. adding an inductance in series with Z

B. adding a capacitance in series with Z

C. adding an inductance in shunt across Z

D. adding a capacitance in shunt across Z

Q3.04.3 MCQ 2004 2M

Consider a 300 Ω, quarter wave long (at 1 GHz) transmission line as shown in figure. It is connected to a 10 V, 50 Ω source at one end and is left open-circuited at the other end. The magnitude of the voltage at the open-circuit end of the line is,

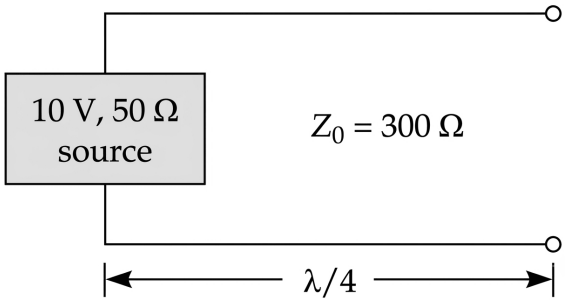


Fig. Q3.04.3

A. 10 V **B.** 5 V
C. 60 V **D.** 60/7 V

Q3.03.1 MCQ 2003 2M

A short-circuited stub is shunt connected to a transmission line as shown in the figure. If $Z_0 = 50\,\Omega$, the admittance Y seen at the junction of the stub and the transmission line is

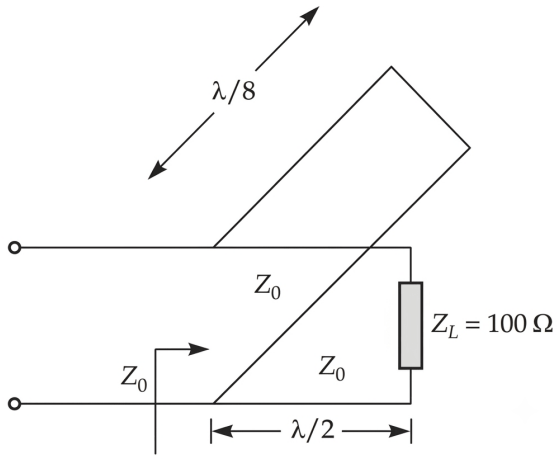


Fig. Q3.03.1

A. (0.01 - j0.02) ℧ **B.** (0.02 - j0.01) ℧
C. (0.04 - j0.02) ℧ **D.** (0.02 + j0) ℧

Q3.02.1 MCQ 2002 1M

In an impedance Smith chart, a clockwise movement along a constant resistance circle gives rise to

A. a decrease in the value of reactance.

B. an increase in the value of reactance.

C. no change in the reactance value.

D. no change in the impedance value.

Q3.02.2 MCQ 2002 1M

The VSWR can have any value between

A. 0 and 1 **B.** -1 and +1
C. 0 and ∞ **D.** 1 and ∞

Q3.01.1 MCQ 2001 2M

A uniform plane electromagnetic wave incident normally on a plane surface of a dielectric material is reflected with a VSWR of 3. What is the percentage of incident power that is reflected?

A. 10% **B.** 25%
C. 50% **D.** 75%

Q3.01.2 MCQ 2001 1M

A transmission line is distortionless if

A. $RL = \dfrac{1}{GC}$ **B.** $RL = GC$

C. $LG = RC$ **D.** $RG = LC$

Q3.00.1 MCQ 2000 1M

The magnitude of the open-circuit and short-circuit input impedance of a transmission line are 100 Ω and 25 Ω respectively. The characteristic impedance of the line is,

A. 25 Ω **B.** 50 Ω

C. 75 Ω **D.** 100 Ω

Q3.99.1 MCQ 1999 2M

In air, a lossless transmission line of length 50 cm with $L = 10\,\mu$H/m, $C = 40$ pF/m is operated at 25 MHz. Its electrical path length is

A. 0.5 meters **B.** λ meters

C. $\pi/2$ radians **D.** 180 degrees

Q3.99.2 MCQ 1999 2M

In a twin-wire transmission line in air, the adjacent voltage maxima are at 12.5 cm and 27.5 cm. The operating frequency is

A. 300 MHz **B.** 1 GHz

C. 2 GHz **D.** 6.28 GHz

Q3.98.1 MCQ 1998 1M

All transmission line section in figure, have a characteristic impedance $R_0 + j0$. The input impedance Z_{in} equals,

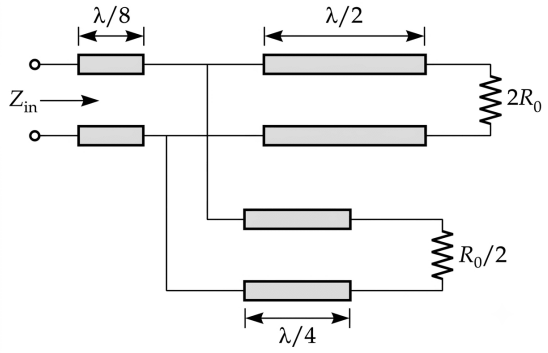


Fig. Q3.98.1

A. $\dfrac{2}{3}R_0$ **B.** R_0 **C.** $\dfrac{3}{2}R_0$ **D.** $2R_0$

Q3.97.1 MCQ 1997 1M

A transmission line of 50 Ω characteristic impedance is terminated with a 100 Ω resistance. The minimum impedance measured on the line is equal to

A. 0 Ω **B.** 25 Ω

C. 50 Ω **D.** 100 Ω

Q3.96.1 MCQ 1996 1M

The capacitance per unit length and the characteristic impedance of a lossless transmission line are C and Z_0 respectively. The velocity of a travelling wave on the transmission line is

A. Z_0C **B.** $\dfrac{1}{(Z_0C)}$

C. $\dfrac{Z_0}{C}$ **D.** $\dfrac{C}{Z_0}$

Q3.96.2 MCQ 1996 1M

A lossless transmission line having 50 Ω characteristic impedance and length $\lambda/4$ is short-circuited at one end and connected to an ideal voltage source of 1 V at the other end. The current drawn from the voltage sources is

A. 0 **B.** 0.02 A

C. ∞ **D.** None of these

Q3.94.1 MCQ 1994 2M

If a pure resistance load, when connected to a lossless 75 Ω line, produces a VSWR of 3 on the line, then the load impedance can only be 25 Ω.

A. True

B. False

Q3.94.2 NAT 1994 1M

A load impedance, $(200 + j0)$ Ω is to be matched to a 50 Ω lossless transmission line by using a quarter wave line transformer (QWT). The characteristic impedance of the QWT required is _____.

Q3.93.1 MCQ 1993 2M

Consider a transmission line of characteristic impedance 50 Ω. Let it be terminated at one end by $(+j50)$ Ω. The VSWR produced by it in the transmission line will be

A. +1 **B.** 0

C. ∞ **D.** $+j$

Q3.92.1 MSQ 1992 2M

A transmission line whose characteristic impedance is a real

- **A.** must be a lossless line.
- **B.** must be a distortionless line.
- **C.** may not be a lossless line.
- **D.** may not be a distortionless line.

Q3.91.1 MCQ 1991 2M

The input impedance of a short-circuited lossless transmission line quarter wave long is

A. purely reactive

B. purely resistive

C. infinite

D. dependent on the characteristic impedance of the line

Q3.89.1 MCQ 1989 2M

A 50 Ω lossless transmission line has a pure reactance of $(j100)$ Ω as its load. The VSWR in the line is

A. 1/2 (Half) **B.** 2 (Two)
C. 4 (Four) **D.** ∞ (Infinity)

Q3.88.1 MCQ 1988 2M

A two-wire transmission line of characteristic impedance Z_0 is connected to a load of impedance $Z_L (Z_L \neq Z_0)$. Impedance matching cannot be achieved with

A. a quarter-wavelength transformer

B. a half-wavelength transformer

C. an open-circuited parallel stub

D. a short-circuited parallel stub

Q3.87.1 MCQ 1987 2M

A transmission line of real characteristic impedance is terminated with an unknown load. The measured value of VSWR on the line is equal to 2 and a voltage minimum point is found to be at the load. The load impedance is then

A. complex **B.** purely capacitive
C. purely resistive **D.** purely inductive

Answer Key

Note: * descriptive/evidence-based, no single definitive answer.
MTA Marks To All awarded — see detailed solutions.

Q3.26.1 B	Q3.26.2 20.38	Q3.25.1 6.4
Q3.24.1 200	Q3.24.2 D	Q3.23.1 0.13
Q3.23.2 B,C,D	Q3.23.3 B,C	Q3.23.4 A
Q3.21.1 D	Q3.20.1 D	Q3.20.2 6.25
Q3.18.1 0.001	Q3.18.2 C	Q3.17.1 0.375
Q3.17.2 49 to 50	Q3.17.3 0.85 to 0.88	Q3.16.1 32.40 to 32.80

Q3.16.2 0.1	Q3.16.3 *	Q3.16.4 B
Q3.15.1 30	Q3.15.2 120.22	Q3.15.3 D
Q3.15.4 B	Q3.14.1 B	Q3.14.2 C
Q3.14.3 33.33	Q3.14.4 70.71	Q3.14.5 40
Q3.14.6 B	Q3.13.1 A	Q3.12.1 C
Q3.12.2 MTA	Q3.11.1 A	Q3.11.2 C

Q3.10.1 C	Q3.10.2 B	Q3.10.3 D
Q3.09.1 D	Q3.08.1 D	Q3.08.2 D
Q3.07.1 C	Q3.07.2 D	Q3.06.1 D
Q3.06.2 B	Q3.05.1 B	Q3.05.2 C
Q3.05.3 A	Q3.05.4 D	Q3.04.1 D
Q3.04.2 A	Q3.04.3 C	Q3.03.1 A

Q3.02.1 B	Q3.02.2 D	Q3.01.1 B
Q3.01.2 C	Q3.00.1 B	Q3.99.1 C
Q3.99.2 B	Q3.98.1 B	Q3.97.1 B
Q3.96.1 B	Q3.96.2 A	Q3.94.1 B
Q3.94.2 100	Q3.93.1 C	Q3.92.1 B,C
Q3.91.1 C	Q3.89.1 D	Q3.88.1 B
Q3.87.1 C		

PrepFusion

UNIT

IV

Rectangular & Circular Waveguides

Topics

1. TE & TM Modes
2. Waveguide Parameters & Power Calculations
3. Field Patterns

PrepFusion

Unit Snapshot*

Stream: EC	**Questions:** 35	**MCQ:** 21	**MSQ:** 5	**NAT:** 9
1M: 11	**2M:** 24	**Descriptive:** 0	**Avg Weightage** ~2M	**Range:** 1M(2026)–4M(2022)

**See preface for how Avg Weightage and Range are calculated.*

Q4.26.1 NAT 2026 1M

The cutoff frequency (in GHz) for the dominant TE_{10} mode of an air-filled rectangular waveguide of inner dimension 0.28 inch × 0.14 inch is _____ (rounded off to two decimal places).

Q4.23.1 MCQ 2023 1M

Consider a narrow band signal, propagating in a lossless dielectric medium ($\epsilon_r = 4, \mu_r = 1$), with phase velocity v_p and group velocity v_g. Which of the following statement is true? (c is the velocity of light in vacuum.)

A. $v_p > c, v_g > c$
B. $v_p < c, v_g > c$
C. $v_p > c, v_g < c$
D. $v_p < c, v_g < c$

Q4.22.1 MSQ 2022 2M

A waveguide consists of two infinite parallel plates (perfect conductors) at a separation of 10^{-4} cm, with air as the dielectric. Assume the speed of light in air to be 3×10^8 m/s. The frequency/frequencies of TM waves which can propagate in this waveguide is/are _____.

☐ **A.** 6×10^{15} Hz

☐ **B.** 0.5×10^{12} Hz

☐ **C.** 8×10^{14} Hz

☐ **D.** 1×10^{13} Hz

Q4.22.2 MSQ 2022 2M

Consider the following wave equation:

$$\frac{\partial^2 f(x,t)}{\partial t^2} = 10000 \frac{\partial^2 f(x,t)}{\partial x^2}$$

Which of the given options is/are solution(s) to the given wave equation?

☐ **A.** $f(x,t) = e^{-(x-100t)^2} + e^{-(x+100t)^2}$

☐ **B.** $f(x,t) = e^{-(x-100t)} + 0.5e^{-(x+1000t)}$

☐ **C.** $f(x,t) = e^{-(x-100t)} + \sin(x + 100t)$

☐ **D.** $f(x,t) = e^{-100\pi(-100x+t)} + e^{j100\pi(100x+t)}$

Q4.21.1 NAT 2021 2M

A standard air-filled rectangular waveguide with dimensions, $a = 8$ cm, $b = 4$ cm, operates at 3.4 GHz. For the dominant mode of wave propagation, the phase velocity of the signal is v_p. The value (Rounded off to two decimal places) of v_p/c, where 'c' denotes the velocity of light, is _____.

Q4.19.1 MCQ 2019 2M

The dispersion equation of a waveguide, which relates the wave number 'k' to the frequency ω, is $k(\omega) = \left(\frac{1}{c}\right)\sqrt{\omega^2 - \omega_0^2}$ where the speed of light $c = 3 \times 10^8$ m/s, and ω_0 is a constant. If the group velocity is 2×10^8 m/s, then the phase velocity is

A. 2×10^8 m/s
B. 3×10^8 m/s
C. 1.5×10^8 m/s
D. 4.5×10^8 m/s

Q4.19.2 NAT 2019 2M

A rectangular waveguide of width 'w' and height 'h' has cut-off frequencies for TE_{10} and TE_{11} modes in the ratio 1 : 2. The aspect ratio w/h, rounded off to two decimal places, is ____.

Q4.18.1 NAT 2018 2M

The cut-off frequency of TE_{01} mode of an air filled rectangular waveguide having inner dimensions a cm × b cm ($a > b$) is twice that of the dominant TE_{10} mode. When the waveguide is operated at a frequency which is 25% higher than the cut-off frequency of the dominant mode, the guide wavelength is found to be 4 cm. The value of b (in cm) correct to two decimal places, is ____.

Q4.17.1 MCQ 2017 2M

Standard air filled rectangular waveguides of dimensions $a = 2.29$ cm and $b = 1.02$ cm are designed for radar applications. It is desired that these waveguides operate only in the dominant TE_{10} mode with the operating frequency at least 25% above the cut-off frequency of the TE_{10} mode but not higher than 95% of the next higher cut-off frequency. The range of the allowable operating frequency, f is ____.

A. 8.19 GHz ≤ f ≤ 13.1 GHz
B. 8.19 GHz ≤ f ≤ 12.45 GHz
C. 6.55 GHz ≤ f ≤ 13.1 GHz
D. 1.64 GHz ≤ f ≤ 10.24 GHz

Q4.16.1 MCQ 2016 2M

Consider an air filled rectangular waveguide with dimensions, $a = 2.286$ and $b = 1.016$ cm. The increasing order of the cut-off frequencies for different modes is

A. $TE_{01} < TE_{10} < TE_{11} < TE_{20}$
B. $TE_{20} < TE_{11} < TE_{10} < TE_{01}$
C. $TE_{10} < TE_{20} < TE_{01} < TE_{11}$

D. $TE_{10} < TE_{11} < TE_{20} < TE_{01}$

Q4.16.2 NAT 2016 2M

Consider an air filled rectangular waveguide with dimensions, $a = 2.286$ cm and $b = 1.016$ cm. At 10 GHz operating frequency, the value of the propagation constant (per meter) of the corresponding propagating mode is ____.

Q4.15.1 NAT 2015 2M

An air filled rectangular waveguide of interval dimensions a cm $\times$ b cm ($a > b$) has a cut-off frequency of 6 GHz for the dominant TE_{10} mode. For the same waveguide, if the cut-off frequency of the TM_{11} mode is 15 GHz, the cut-off frequency of the TE_{01} mode (in GHz) is ____.

Q4.15.2 MCQ 2015 2M

The longitudinal component of the magnetic field inside an air filled rectangular waveguide made of a perfect electric conductor is given by the following expression: $H_z(x, y, z, t) = 0.1\cos(25\pi x)\cos(30.3\pi y)\cos(12\pi \times 10^9 t - \beta z)$ A/m The cross-sectional dimensions of the waveguide are given as $a = 0.08$ m and $b = 0.033$ m. The mode of propagation inside the waveguide is

A. TM_{12} **B.** TM_{21}
C. TE_{21} **D.** TE_{12}

Q4.14.1 NAT 2014 2M

Consider an air filled rectangular waveguide with a cross-section of 5 cm $\times$ 3 cm. For this waveguide, the cut-off frequency (in MHz) of TE_{21} mode is ____.

Q4.14.2 NAT 2014 2M

For a rectangular waveguide of internal dimensions $a \times b$ ($a > b$), the cut-off frequency for the TE_{11} mode is the arithmetic mean of the cut-off frequencies for TE_{10} mode and TE_{20} mode. If $a = \sqrt{5}$ cm, the value of b (in cm) is ____.

Q4.12.1 MCQ 2012 2M

The magnetic field along the propagation direction inside a rectangular waveguide with the cross-section shown in figure is

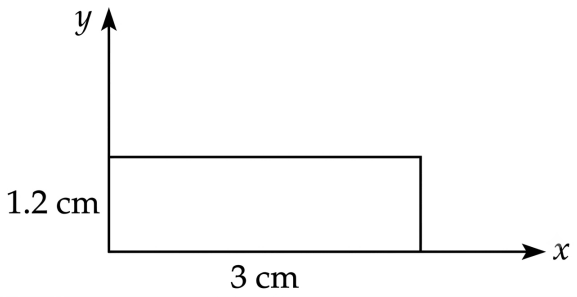


Fig. Q4.12.1

$H_z = 3\cos(2.094 \times 10^2 x)\cos(2.618 \times 10^2 y)\cos(6.283 \times 10^{10} t - \beta z)$ The phase velocity v_p of the wave inside the waveguide satisfies

A. $v_p > c$ **B.** $v_p = c$
C. $0 < v_p < c$ **D.** $v_p = 0$

Q4.11.1 MCQ 2011 1M

The modes in a rectangular waveguide are denoted by TE_{mn}/TM_{mn} where 'm' and 'n' are the eigen numbers along the larger and smaller dimensions of the waveguide respectively. Which one of the following statements is true?

A. The TM_{10} mode of the waveguide does not exist.

B. The TE_{10} mode of the waveguide does not exist.

C. The TM_{10} and the TE_{10} modes both exist and have the same cut-off frequencies.

D. The TM_{10} and the TE_{01} modes both exist and have the same cut-off frequencies.

Q4.09.1 MCQ 2009 1M

Which of the following statements is true regarding the fundamental mode of the metallic waveguides shown?

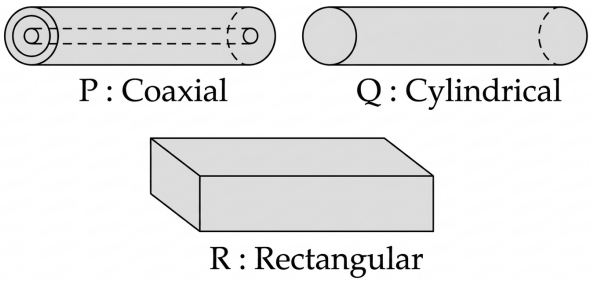


Fig. Q4.09.1

A. Only P has no cut-off frequency.

B. Only Q has no cut-off frequency.

C. Only R has no cut-off frequency.

D. All three have cut-off frequencies.

Q4.08.1 MCQ 2008 2M

A rectangular waveguide of internal dimensions (a = 4 cm and b = 3 cm) is to be operated in TE_{11} mode. The minimum operating frequency is

A. 6.25 GHz **B.** 6.0 GHz
C. 5.0 GHz **D.** 3.75 GHz

Q4.07.1 MCQ 2007 2M

The $\vec{E}$ field in a rectangular waveguide of inner dimensions $a \times b$ is given by

$$\vec{E} = \frac{\omega\mu}{h^2}\left(\frac{\pi}{a}\right) H_0 \sin\left(\frac{2\pi x}{a}\right) \sin(\omega t - \beta z)\,\hat{y}$$

where H_0 is a constant, and a and b are the dimensions along the x-axis and the y-axis respectively. The mode of propagation in the waveguide is.

A. TE_{20} **B.** TE_{11}
C. TM_{20} **D.** TE_{10}

Q4.07.2 MCQ 2007 2M

An air filled rectangular waveguide has inner dimensions of 3 cm × 2 cm. The wave impedance of the TE_{20} mode of propagation in the waveguide at a frequency of 30 GHz is (free space impedance η_0 = 377 Ω)

A. 308 Ω **B.** 355 Ω
C. 400 Ω **D.** 461 Ω

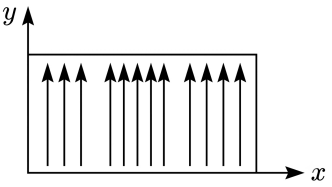
Q4.06.1 MCQ 2006 2M

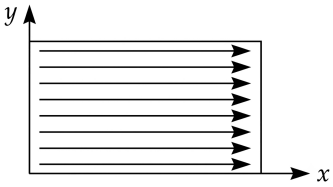
A rectangular waveguide having TE_{10} mode as dominant mode is having a cut-off frequency of 18 GHz for the TE_{30} mode. The inner broad wall dimension of the rectangular waveguide is

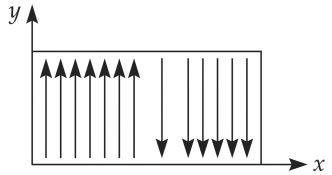
A. 5/3 cm **B.** 5 cm
C. 5/2 cm **D.** 10 cm

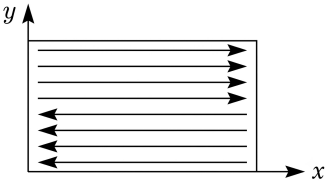
Q4.05.1 MCQ 2005 2M

Which one of the following does represent the electric field lines for the TE_{02} mode in the cross-section of a hollow rectangular metallic waveguide?

A.


B.


C.


D.


Q4.04.1 MCQ 2004 1M

The phase velocity of an electromagnetic wave propagating in a hollow metallic rectangular waveguide in the TE_{10} mode is

A. equal to its group velocity.

B. less than the velocity of light in free space.

C. equal to the velocity of light in free space.

D. greater than the velocity of light in free space.

Q4.03.1 MCQ 2003 2M

A rectangular metal wave guide filled with a dielectric material of relative permittivity ϵ_r = 4 has the inside dimensions 3.0 cm × 1.2 cm. The cut-off frequency for the dominant mode is

A. 2.5 GHz **B.** 5.0 GHz
C. 10.0 GHz **D.** 12.5 GHz

Q4.02.1 MCQ 2002 1M

The phase velocity for the TE_{10} mode in an air filled rectangular waveguide is Note: c is the velocity of plane waves in free space.

A. less than c **B.** equal to c
C. greater than c **D.** none of the above

Q4.01.1 MCQ 2001 1M

The dominant mode in a rectangular waveguide is TE_{10}, because this mode has

A. no attenuation

B. no cut-off

C. no magnetic field component

D. the highest cut-off wavelength

Q4.01.2 MCQ 2001 1M

The phase velocity of waves propagating in a hollow metal waveguide is

A. greater than the velocity of light in free space.

B. less than the velocity of light in free space.

C. equal to the velocity of light in free space.

D. equal to the group velocity.

Q4.00.1 MCQ 2000 2M

A rectangular waveguide has dimensions 1 cm × 0.5 cm. Its cut-off frequency is

A. 5 GHz **B.** 10 GHz
C. 15 GHz **D.** 12 GHz

Q4.99.1 MCQ 1999 1M

Assuming perfect conductors of a transmission line, pure TEM propagation is not possible in

A. coaxial cable

B. air filled cylindrical waveguide

C. parallel twin-wire line in air

D. semi-infinite parallel to plate waveguide

Q4.97.1 MCQ 1997 1M

A rectangular air filled waveguide has a cross-section of 4 cm × 10 cm. The minimum frequency which can propagate in the waveguide is

A. 1.5 GHz **B.** 2.0 GHz
C. 2.5 GHz **D.** 3.0 GHz

Q4.94.1 NAT 1994 1M

The interior of a $\frac{20}{3}$ cm × $\frac{20}{4}$ cm rectangular waveguide is completely filled with a dielectric of $\epsilon_r = 4$. Waves of free space wave lengths shorter than _____ can be propagated in the TE_{11} mode.

Q4.90.1 MSQ 1990 2M

Choose the correct statements: For a wave propagating in an air filled rectangular waveguide

☐ **A.** guided wavelength is never less than the free space wavelength.

☐ **B.** wave impedance is never less than the free space impedance.

☐ **C.** phase velocity is never less than the free space velocity.

☐ **D.** TEM mode is possible if the dimensions of the wave guide are properly chosen.

Q4.88.1 MSQ 1988 2M

For a normal mode EM wave propagating in a hollow rectangular wave guide

☐ **A.** the phase velocity is greater than the group velocity.

☐ **B.** the phase velocity is greater than velocity of light in free space.

☐ **C.** the phase velocity is less than the velocity of light in free space.

☐ **D.** the phase velocity may be either greater than or less than the group velocity.

Q4.87.1 MSQ 1987 2M

The cut-off frequency of a waveguide depends upon

☐ **A.** the dimensions of waveguide.

☐ **B.** the dielectric property of the medium in the waveguide.

☐ **C.** the characteristic impedance of the waveguide.

☐ **D.** the transverse and axial components of the fields.

Answer Key

Q4.26.1 21.09	**Q4.23.1** D	**Q4.22.1** A,C
Q4.22.2 A,C	**Q4.21.1** 1.198	**Q4.19.1** D
Q4.19.2 1.72 to 1.74	**Q4.18.1** 0.75	**Q4.17.1** B
Q4.16.1 C	**Q4.16.2** 157.5 to 158.5	**Q4.15.1** 13.70 to 13.80
Q4.15.2 C	**Q4.14.1** 7800 to 7820	**Q4.14.2** 2
Q4.12.1 D	**Q4.11.1** A	**Q4.09.1** A
Q4.08.1 A	**Q4.07.1** A	**Q4.07.2** C
Q4.06.1 C	**Q4.05.1** D	**Q4.04.1** D
Q4.03.1 A	**Q4.02.1** C	**Q4.01.1** D
Q4.01.2 A	**Q4.00.1** C	**Q4.99.1** B
Q4.97.1 A	**Q4.94.1** 8	**Q4.90.1** A,C
Q4.88.1 A,B	**Q4.87.1** A,B	

UNIT

V

Antenna & Antenna Array

Topics

1. Types of Antenna & Their Radiation Patterns
2. Antenna Parameters
3. Friis Equations
4. Antenna Arrays & Their Radiation Patterns

PrepFusion

Unit Snapshot*

Stream: EC	**Questions:** 46	**MCQ:** 38	**MSQ:** 1	**NAT:** 7
1M: 19	**2M:** 27	**Descriptive:** 0	**Avg Weightage** ~2M	**Range:** 1M(2026)–3M(2016)

**See preface for how Avg Weightage and Range are calculated.*

Q5.26.1 MCQ 2026 1M

Consider the Friis' transmission equation,

$$P_R = \frac{P_T G_T G_R \lambda^2}{(4\pi D)^2}$$

where P_R and P_T are the received and the transmitted powers, respectively. G_T and G_R are the gain of transmitting and receiving antennas, respectively, D is the distance between the transmitting and receiving antennas, and λ is the wavelength in free space. Given: $G_T = G_R = 1.0$, $\lambda = 0.30$ m and $P_T = +10$ dBm. Choose the distance (D), in km, from the following options at which the received power, $P_R = -90$ dBm?

A. $\frac{15}{4\pi}$ **B.** $\frac{15}{2\pi}$
C. $\frac{75}{2\pi}$ **D.** $\frac{3}{4\pi}$

Q5.21.1 NAT 2021 2M

An antenna with a directive gain of 6 dB is radiating a total power of 16 kW. The amplitude of the electric field in free space at a distance of 8 km from the antenna in the direction of 6 dB gain (Rounded off to 3 decimal places) is _____ V/m.

Q5.20.1 MCQ 2020 2M

For an infinitesimally small dipole in free space, the electric field E_θ in the far field is proportional to $(\frac{e^{-jkr}}{r})\sin\theta$, where $k = 2\pi/\lambda$. A vertical infinitesimally small electric dipole ($\delta I \ll \lambda$) is placed at a distance $h (h > 0)$ above an infinite ideal conducting plane, as shown in the figure. The minimum value of h, for which one of the maxima in the far field radiation pattern occurs at $\theta = 60°$, is

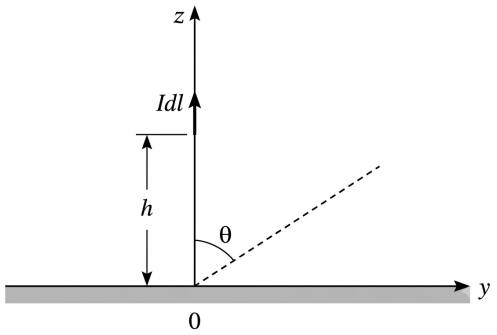


Fig. Q5.20.1

A. 0.75λ **B.** λ
C. 0.25λ **D.** 0.5λ

Q5.19.1 NAT 2019 1M

Radiation resistance of a small dipole current element of length l at a frequency of 3 GHz is 3 Ω. If the length is changed by 1%, then the percentage change in the radiation resistance, rounded off to two decimal places, is _____ %.

Q5.17.1 MCQ 2017 2M

A half wavelength dipole is kept in the x-y plane and oriented along 45° from the x-axis. Determine the direction of null in the radiation pattern for $0 \le \phi \le \pi$. Here the angle $\theta(0 \le \theta \le \pi)$ is measured from the z-axis, and the angle $\phi(0 \le \theta \le 2\pi)$ is measured from the x-axis in the x-y plane.

A. $\theta = 90°, \phi = 45°$ **B.** $\theta = 45°, \phi = 90°$
C. $\theta = 90°, \phi = 135°$ **D.** $\theta = 45°, \phi = 135°$

Q5.17.2 MCQ 2017 1M

Consider a wireless communication link between a transmitter and a receiver located in free space, with finite and strictly positive capacity. If the effective areas of the transmitter and the receiver antennas, and the distance between them are all doubled, and everything else remains unchanged, the maximum capacity of the wireless link

A. increased by a factor of 2

B. decreased by a factor of 2

C. remains unchanged

D. decreases by a factor of $\sqrt{2}$

Q5.16.1 MCQ 2016 2M

The far zone power density radiated by a helical antenna is approximated as,

$$\vec{W}_{\text{rad}} = \vec{W}_{\text{average}} \approx \hat{a}_r C_0 \frac{1}{r^2}\cos^4\theta$$

The radiated power density is symmetrical with respect to ϕ and exists only in the upper hemisphere $0 \le \theta \le \frac{\pi}{2}, 0 \le \phi \le 2\pi$, C_0 is a constant. The power radiated by the antenna (in Watts) and the maximum directivity of antenna, respectively, are

A. $15C_0$, 10 dB **B.** $1.256C_0$, 10 dB
C. $1.256C_0$, 12 dB **D.** $1.5C_0$, 12 dB

Q5.16.2 NAT 2016 2M

A radar operating at 5 GHz uses a common antenna for transmission and reception. The antenna has a gain of 150 and a aligned for maximum directional radiation and reception to a target 1 km away having radar cross-section of 3 m^2. If it transmits 100 kW, then the received power (in μW) is _____.

Q5.16.3 MCQ 2016 2M

The electric field of a uniform plane wave travelling along the negative z-direction is given by the following equation:

$$\vec{E}_w^i = (\hat{a}_x + j\hat{a}_y)E_0 e^{jkz}$$

This wave is incident upon a receiving antenna placed at the origin and whose radiated electric field towards the incident wave is given by the following equation:

$$\vec{E}_a = (\hat{a}_x + 2\hat{a}_y)E_1 \frac{1}{r} e^{-jkr}$$

The polarization of the incident wave, the polarization of the antenna and losses due to polarization mismatch are, respectively,

A. linear, circular (clockwise), −5 dB

B. circular (clockwise), linear, −5 dB

C. circular (clockwise), linear, −3 dB

D. circular (anticlockwise), linear, −3 dB

Q5.16.4 NAT 2016 2M

Two lossless X-band horn antennas are separated by a distance of 200λ. The amplitude reflection coefficients at the terminals of the transmitting and receiving antennas are 0.15 and 0.18, respectively. The maximum directivities of the transmitting and receiving antennas (over the isotropic antenna) are 18 dB and 22 dB, respectively. Assuming that the input power in the lossless transmission line connected to the antenna is 2 W, and that the antennas are perfectly aligned and polarization matched, the power (in mW) delivered to the load at the receiver is _____.

Q5.15.1 MCQ 2015 1M

The directivity of an antenna array can be increased by adding more antenna elements, as a larger number of elements

A. improves the radiation efficiency

B. increases the effective area of the antenna

C. results in a better impedance matching

D. allows more power to be transmitted by the antenna

Q5.15.2 NAT 2015 2M

Two half-wave dipole antennas placed as shown in the figure are excited with sinusoidally varying currents of frequency 3 MHz and phase shift of $\pi/2$ between them (the element at the origin leads in phase). If the maximum radiated E field at the point P in the x-y plane occurs at an azimuthal angle of 60°, the distance d (in meters) between the antenna is _____.

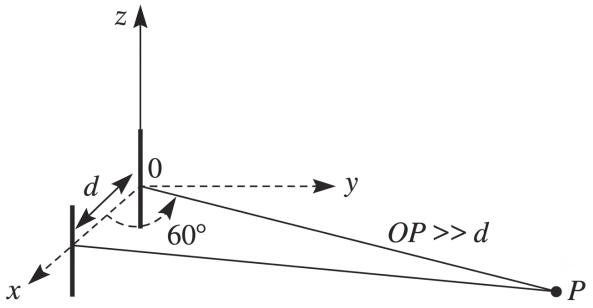


Fig. Q5.15.2

Q5.14.1 MCQ 2014 1M

Match Column-A with Column-B.

Column-A	Column-B
1. Point electromagnetic source	**P.** Highly directional
2. Dish antenna	**Q.** End fire
3. Yagi-Uda antenna	**R.** Isotropic

A. 1 - P, 2 - Q, 3 - R

B. 1 - R, 2 - P, 3 - Q

C. 1 - Q, 2 - P, 3 - R

D. 1 - R, 2 - Q, 3 - P

Q5.14.2 NAT 2014 1M

For an antenna radiating in free space, the electric field at a distance of 1 km is found to be 12 mV/m. Given the intrinsic impedance of the free space is 120$\pi\Omega$, the magnitude of average power density due to this antenna at a distance of 2 km from the antenna (in nW/m^2) is _____.

Q5.14.3 NAT 2014 2M

In spherical coordinates, let $\hat{a}_\theta, \hat{a}_\phi$ denote unit vectors along the θ, ϕ-directions.

$$\vec{E} = \frac{100}{r}\sin\theta\cos(\omega t - \beta r)\hat{a}_\theta \text{ V/m}$$

$$\vec{H} = \frac{0.265}{r}\sin\theta\cos(\omega t - \beta r)\hat{a}_\phi \text{ V/m}$$

represent the electric and magnetic field components of the EM wave at large distances r from a dipole antenna, in free space. The average power (W) crossing the hemispherical shell located at $r = 1$ km, $0 \le \theta \le \pi/2$ is _____.

Q5.12.1 MCQ 2012 1M

The radiation pattern of an antenna in spherical co-ordinates is given by

$$F(\theta) = \cos^4\theta, \quad 0 \le \theta \le \frac{\pi}{2}$$

The directivity of the antenna is

A. 10 dB **B.** 12.6 dB
C. 11.5 dB **D.** 18 dB

Q5.11.1 MCQ 2011 1M

Consider the following statements regarding the complex pointing vector $\vec{P}$ for the power radiated by a point source in an infinite homogenous and lossless medium. $\text{Re}(\vec{P})$ denotes the real part of $\vec{P}$, S denotes a spherical surface whose center is at the point source, and $\hat{n}$ denotes the unit surface normal on S. Which of the following statements is true?

A. $\text{Re}(\vec{P})$ remains constant at any radial distance from the source.

B. $\text{Re}(\vec{P})$ increases with increasing radial distance from the source.

C. $\iint_S \text{Re}(\vec{P})\cdot\hat{n}\,dS$ remains constant at any radial distance from the source.

D. $\iint_S \text{Re}(\vec{P})\cdot\hat{n}\,dS$ decreases with increasing radial distance from the source.

Q5.08.1 MCQ 2008 2M

At 20 GHz, the gain of a parabolic dish antenna of 1 meter diameter and 70% efficiency is

A. 15 dB **B.** 25 dB
C. 35 dB **D.** 45 dB

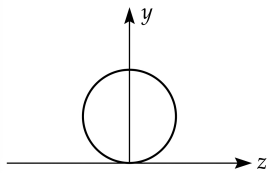
Q5.08.2 MCQ 2008 1M

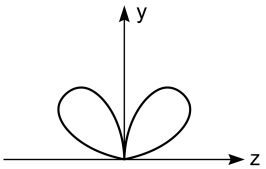
For a Hertz dipole antenna, the Half Power Beam Width (HPBW) in the E-plane is

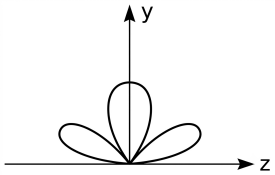
A. 360° **B.** 180° **C.** 90° **D.** 45°

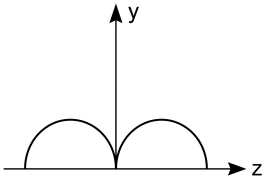
Q5.07.1 MCQ 2007 2M

A $\lambda/2$ dipole is kept horizontally at a height of $\lambda_0/2$ above a perfectly conducting infinite ground plane. The radiation pattern in the plane of the dipole ($\vec{E}$ plane) looks approximately as,

A.


B.


C.


D.


Q5.06.1 MCQ 2006 2M

A mast antenna consisting of a 50 meter long vertical conductor operates over a perfectly conducting ground plane. It is base fed at a frequency of 600 kHz. The radiation resistance of the antenna (in Ω) is

A. $\frac{2\pi^2}{5}$ **B.** $\frac{\pi^2}{5}$
C. $\frac{4\pi^2}{5}$ **D.** $20\pi^2$

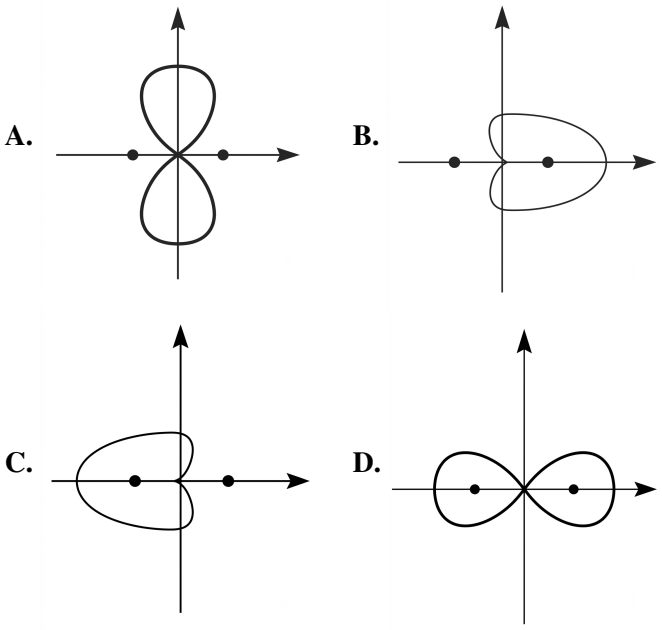
Q5.06.2 MCQ 2006 1M

A transmission line is feeding 1 W of power to a horn antenna having gain of 10 dB. The antenna is matched to the transmission line. The total power radiated

A. 10 Watts **B.** 1 Watt
C. 0.1 Watt **D.** 0.01 Watt

Q5.05.1 MCQ 2005 2M

Two identical and parallel dipole antennas are kept apart by a distance of $\lambda/4$ in the H-plane. They are fed with equal currents but the right most antenna has a phase shift of +90°. The radiation pattern is given as,



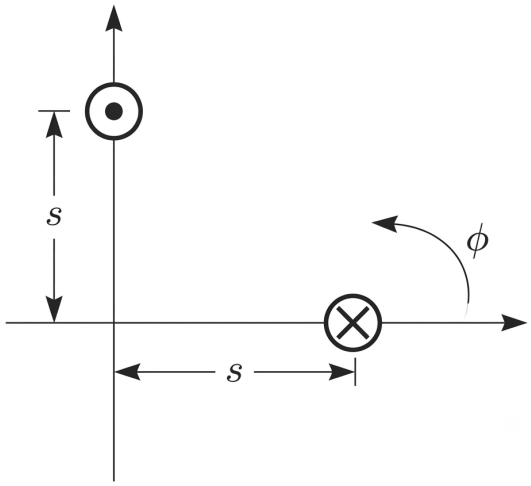


Fig. Q5.03.1

Q5.04.1 MCQ 2004 1M

Consider a lossless antenna with a directive gain of +6 dB. If 1 mW of power is fed to it the total power radiated by the antenna will be

A. 4 mW **B.** 1 mW
C. 7 mW **D.** $\frac{1}{4}$ mW

Q5.03.1 MCQ 2003 2M

Two identical antennas are placed in the $\theta = \pi/2$ plane as shown in the figure. The elements have equal amplitude excitation with 180° polarity difference, operating at wavelength λ. The correct values of the magnitude of the far zone resultant electric field strength normalized with that of a single element, both computed for $\phi = 0$, is

A. $2\cos\left(\frac{2\pi s}{\lambda}\right)$ **B.** $2\sin\left(\frac{2\pi s}{\lambda}\right)$
C. $2\cos\left(\frac{\pi s}{\lambda}\right)$ **D.** $2\sin\left(\frac{\pi s}{\lambda}\right)$

Q5.02.1 MCQ 2002 2M

A person with a receiver is 5 km away from the transmitter. What is the distance that this person must move further to detect a 3 dB decrease in signal strength?

A. 942 m **B.** 2070 m
C. 4978 m **D.** 5320 m

Q5.02.2 MCQ 2002 1M

The line of sight communication requires the transmit and receive antennas to face each other. If the transmit antenna is vertically polarized, for best reception the receiver antenna should be

A. horizontally polarized

B. vertically polarized

C. at 45° with respect to horizontal polarized

D. at 45° with respect to vertical polarized

Q5.01.1 MCQ 2001 2M

In a uniform linear array, four isotropic radiating elements are spaced $\lambda/4$ apart. The progressive phase shift between the elements required for forming the main beam at 60° off the end fire is

A. $-\pi$ radians **B.** $-\pi/2$ radians
C. $-\pi/4$ radians **D.** $-\pi/8$ radians

Q5.01.2 MCQ 2001 2M

A medium wave radio transmitter operating at a wavelength of 492 m has a tower antenna of height 124 m. What is the radiation resistance of the antenna?

A. 25Ω **B.** 36.5Ω
C. 50Ω **D.** 73Ω

Q5.00.1 MCQ 2000 2M

For an 8 feet (2.4 m) parabolic dish antenna operating at 4 GHz, the minimum distance required for far field measurement is closest to

A. 7.5 cm **B.** 15 cm
C. 15 m **D.** 150 m

Q5.00.2 MCQ 2000 1M

If the diameter of a $\lambda/2$ dipole antenna is increased from $\frac{\lambda}{100}$ to $\frac{\lambda}{50}$, then its

A. bandwidth increases **B.** bandwidth decreases
C. gain increases **D.** gain decreases

Q5.99.1 MCQ 1999 2M

A transmitting antenna radiates 251 W isotropically. A receiving antenna, located 100 m away from the transmitting antenna, has an effective aperture of 500 cm^2. The total power received by the antenna is

A. 10 μW **B.** 1 μW
C. 20 μW **D.** 100 μW

Q5.98.1 MCQ 1998 1M

The far field on an antenna varies with distance 'r' as,

A. $\frac{1}{r}$ **B.** $\frac{1}{r^2}$
C. $\frac{1}{r^3}$ **D.** $\frac{1}{\sqrt{r}}$

Q5.98.2 MCQ 1998 1M

An antenna in free space receives 2 μW of power when the incident electric field is 20 mV/m rms. The effective aperture of the antenna is

A. 0.005 m^2 **B.** 0.05 m^2
C. 1.885 m^2 **D.** 3.77 m^2

Q5.98.3 MCQ 1998 1M

The radiation resistance of a circular loop of one turn is 0.01 Ω. The radiation resistance of five turns of such a loop will be

A. 0.002 Ω **B.** 0.01 Ω
C. 0.05 Ω **D.** 0.25 Ω

Q5.98.4 MCQ 1998 1M

The vector $\vec{H}$ in the far field on an antenna satisfies

A. $\nabla \cdot \vec{H} = 0$ and $\nabla \times \vec{H} = 0$

B. $\nabla \cdot \vec{H} \neq 0$ and $\nabla \times \vec{H} \neq 0$

C. $\nabla \cdot \vec{H} = 0$ and $\nabla \times \vec{H} \neq 0$

D. $\nabla \cdot \vec{H} \neq 0$ and $\nabla \times \vec{H} = 0$

Q5.97.1 MCQ 1997 2M

A parabolic dish antenna has conical beam 2° wide. The directivity of the antenna is approximately

A. 20 dB **B.** 30 dB
C. 40 dB **D.** 50 dB

Q5.96.1 MCQ 1996 2M

At 1 km long microwave link use two antennas each having 30 dB gain. If the power transmitted by one antenna is 1 W at 3 GHz, the power received by the other antenna is approximately

A. 98.6 μW **B.** 76.8 μW
C. 63.4 μW **D.** 55.2 μW

Q5.96.2 MCQ 1996 2M

A transverse electromagnetic wave with circular polarization is received by a dipole antenna. Due to polarization mismatch, the power transfer efficiency from the wave to the antenna is reduced to about

A. 50% **B.** 35.3%
C. 25% **D.** 0%

Q5.95.1 MCQ 1995 1M

An antenna, when radiating has a highly directional radiation pattern. When the antenna is receiving, its radiation pattern

A. is more directive **B.** is less directive
C. is the same **D.** exhibits no directivity at all

Q5.94.1 MCQ 1994 1M

For a half-wave dipole antenna, which of the following statements are right?

A. The radiation intensity is maximum along the normal to the dipole axis.

B. The current distribution along its length is uniform irrespective of the length.

C. The effective length equals its physical length.

D. The input impedance is independent of the location of the feed point.

Q5.92.1 MCQ 1992 2M

The beamwidth between first nulls of a uniform linear array of N equally spaced (element spacing = d), equal excited antennas, is determine by

A. N alone and not by d **B.** d alone and not by N
C. the ratio, (N/d) **D.** the product, (Nd)

Q5.92.2 MCQ 1992 2M

Two dissimilar antennas having their maximum directivities equal, which of the following statements are right?

A. must have their beam widths also equal.

B. cannot have their beam widths equal because they are dissimilar antenna.

C. may not necessarily have their maximum power gains equal.

D. must have their effective aperture areas (capture areas) also equal.

Q5.91.1 MCQ 1991 2M

In a broad side array of 20 isotropic radiators, equally spaced at distance of $\lambda/2$, the beam width between first nulls is

A. 51.3 degrees **B.** 11.46 degrees
C. 22.9 degrees **D.** 102.6 degrees

Q5.90.1 MCQ 1990 2M

Two isotropic antennas are separated by a distance of two wavelengths. If both the antennas are fed with currents of equal phase and magnitude, the number of lobes in the radiation pattern in the horizontal plane are

A. 2 **B.** 4
C. 6 **D.** 8

Q5.88.1 MSQ 1988 2M

The electric field E and the magnetic field H of a short dipole antenna satisfy the condition

- ☐ **A.** the r component of E is equal to zero.
- ☐ **B.** both r and θ components of H are equal to zero.
- ☐ **C.** the θ component of E dominates the r component in the far field region.
- ☐ **D.** the θ and ϕ components of H are of the same order of magnitude in the near field region.

Answer Key

Q5.26.1 B	Q5.21.1 0.244	Q5.20.1 B
Q5.19.1 2.01	Q5.17.1 A	Q5.17.2 C
Q5.16.1 B	Q5.16.2 0.012	Q5.16.3 C
Q5.16.4 3	Q5.15.1 B	Q5.15.2 50
Q5.14.1 B	Q5.14.2 47.5 to 48.0	Q5.14.3 55.5
Q5.12.1 A	Q5.11.1 C	Q5.08.1 D
Q5.08.2 C	Q5.07.1 B	Q5.06.1 A
Q5.06.2 B	Q5.05.1 C	Q5.04.1 B
Q5.03.1 D	Q5.02.1 B	Q5.02.2 B
Q5.01.1 C	Q5.01.2 B	Q5.00.1 D
Q5.00.2 C	Q5.99.1 D	Q5.98.1 A
Q5.98.2 C	Q5.98.3 D	Q5.98.4 C
Q5.97.1 C	Q5.96.1 C	Q5.96.2 A
Q5.95.1 C	Q5.94.1 A	Q5.92.1 D
Q5.92.2 C	Q5.91.1 B	Q5.90.1 D
Q5.88.1 B,C		

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